SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Syllabus
Last Year
B. Tech in Computer Engineering
(Programme commenced from AY 2023-24)
(Department of Computer Engineering)

From
Academic Year 2026-27
(SVU-KJSSE 2.0)

(Approved by BOS dated 10th April 2026, FOET dated27th November,2025 and AC dated 08th May, 2026)

Preamble

KJSSE as a constituent college of Somaiya Vidyavihar University has the academic flexibility to develop and implement its own curriculum. *KJSCE-SVU-2020* with features such as inclusion of choice based Open Elective Courses, Audit Courses, Add on Credit Courses, Exposure Courses, etc. Distinct assessment and evaluation methods are also designed based on focus of individual courses. The outcome of this entire exercises; either by way of student placements or the feedback received from all stakeholders is quite encouraging.

At present, Industry is moving towards Industrial revolution 4.0. Knowing very well that every country's education system forms the basis of its progress and the groundwork for its future, we need to be making engineering graduates equipped to take industrial challenges. A common feature in successful education systems is the balance between tradition and the capacity to be flexible and able to adapt to current social trends. To achieve this, Somaiya Vidyavihar University allows for the undergraduate courses to have a focus on the changing industrial scenario.

Our new revision in syllabus *KJSSE-SVU-2023,* to be introduced from the academic year 2023-24, has been designed based on the revised guidelines from various accrediting bodies.

The said syllabus is a result of expert advice from members of Board of Studies, Faculty of Engineering & Technology and Academic Council; both having due representation from academia as well as appropriate industries. Subsequently faculty members of the college have put in efforts to document it in the form which has been presented here.

Some of the highlights of the **KJSCE-SVU-2023** syllabus are: Introduction of wide choice for branch specific electives, more number of open or interdisciplinary electives, opportunity for internships, etc. Competitive Programming course is introduced in SY curriculum paying more attention to hands-on learning.

Focus of academic processes in KJSSE is such that, by the time student completes the requirements of the degree, he/ she will be able to acquire attributes required for profession as an engineer. Outcomes are defined to acquire these attributes which lead to development of curriculum, pedagogy and assessment tools. These tools need to be updated based on experiences of teacher and learner. Hence teaching -learning -evaluation paradigm is going to be a mix of traditional as well as use of ICT tools. Role of the faculty member changes from tutor to trainer / instructor/ facilitator / mentor based on the outcomes targeted.

For measuring learning outcomes of students, traditional methods like tests, laboratory work and End Semester Examinations (ESE) are implemented. Continuous Assessment (CA) is carried out along with laboratory courses, supporting assessment of term-work; through tests and internal assessment (IA) like quizzes, case studies, mini projects etc. These IA tools enable the students to develop competencies through solutions discussed, improvisations suggested, feedbacks given by faculty members. Through these assessment methods students get opportunity for reading research papers, presenting ideas and working in a team. Since the assessments are distributed throughout the term the learning process is continuously monitored and graded.

The Department of Computer Engineering courses focus on thrust areas of Department. These areas are Intelligent System and Data Processing, Network System and Security, Image Analysis and Interpretation and System & Software Engineering.

College promotes co-curricular, extra-curricular activities as well as sports; making life outside classroom exciting and rewarding. What makes these activities very effective is the fact that these do not focus only on winning trophies but try to nurture generic skills such as leadership, effective communication, teamwork etc. which are essential skills for a bright professional career.

Along with my colleagues, I welcome you to Department of Computer Engineering and look forward to lead you towards professional career.

Dr. Suresh Ukarande
Dean, Faculty of Engineering and Technology,
Somaiya Vidyavihar University

Board of Studies in Computer Engineering

Dr. Deepak Sharma	Chairman
Dr. Parikshit Mahalle	Academician Member
Mr. Chetan Mistry	Industry Member
Dr. G. Sivakumar	Research Institute Member
Mr. Dhaval Chothani	Alumni Industry Member
Dr. Manish Potey	Faculty Member (Professor)
Dr. Bhakti Palkar	Faculty Member (Associate Professor)
Dr. Archana Gupta	Faculty Member (Assistant Professor)

Vision

To become a center of excellence in discipline of Computer Engineering for developing technically adept professionals with ethical and leadership qualities in service of society.

Mission

- Provide sound technical foundation in Computer Engineering through comprehensive curriculum and application-oriented learning.
- Provide ambience for professional growth and lifelong learning for adapting to challenges in rapidly changing technology
- Inculcate social and ethical values and leadership qualities

Program Educational Outcomes (PEO)

A graduate of Computer Engineering will

PEO1. Solve problems in diverse fields using knowledge of Computer Engineering.

PEO2. Excel in professional career, exhibit leadership qualities with ethics &soft skills.

PEO3. Pursue higher education, research or entrepreneurship, engage in professional development, adapt to emerging technologies.

Program Outcomes (PO)

After successful completion of the program Computer Engineering Graduate will be able to:

PO 1: ***Engineering knowledge:*** *Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.*

PO 2: ***Problem analysis****: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.*

PO 3: ***Design/development of solutions:*** *Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.*

PO 4: ***Conduct investigations of complex problems:*** *Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.*

PO 5: ***Modern tool usage:*** *Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.*

PO 6: ***The engineer and society:*** *Apply reasoning informed by the contextual knowledge to assess societal, cultural, environmental, health, safety and legal issues relevant to the professional engineering practice; understanding the need of sustainable development*

PO 7: ***Multidisciplinary Competence:*** *Recognize/ study/ analyze/ provide solutions to real-life problems of multidisciplinary nature from diverse fields*

PO 8: ***Ethics:*** *Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice*

PO 9: ***Individual and team work:*** *Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings*

PO 10: ***Communication:*** *Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.*

PO 11: ***Project management and finance:*** *Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.*

PO 12: ***Life-long learning:*** *Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.*

Program Specific Outcomes (PSO)

PSO 1: Design, construct and implement hardware and software based modern Computing / Information systems with varying complexities

PSO 2: Demonstrate competence in designing, implementation and maintenance of computer-based applications, computer-controlled equipment and networks of intelligent devices

Acronyms for Category of course and Syllabus Document

Acronym for category of courses		Acronyms used in syllabus document	
Acronym	**Definition**	**Acronym**	**Definition**
BS	Basic Science Courses	**CA**	Continuous Assessment
ES	Engineering Science	**ESE**	End Semester Exam
HS	Humanities and Social Sciences including Management Courses	**IA**	Internal Assessment
PC	Professional Core Courses	**O**	Oral
PE	Professional Elective courses	**P**	Practical
OE	Open Elective Courses	**P&O**	Practical and Oral
LC	Laboratory Courses	**TH**	Theory
PR	Project	**TUT**	Tutorial
AC	Audit Course	**TW**	Term work
AOCC	Add on Credit Course	**ISE**	In- Semester Examination
AOAC	Add on Audit Course	**CO**	Course Outcome
AVAC	Add on Value Audit Course	**PO**	Program Outcome
EX	Exposure Course	**PSO**	Program specific Outcome
I	Interdisciplinary courses		

Acronyms used for type of Course

Acronym used	Definition

C	Core Course
E	Elective Course
O	Open Elective Technical
H	Open Elective Humanities/Management/SWAYAM-NPTEL
P	Project
L	Laboratory Course
T	Tutorial
X	Exposure course
A	Audit course

Acronyms used in Eight Digit Course code e.g. 216U06C101

Acronym Serially as per code	**Definition**
2	SVU 2023 Second revision
16	College code
U	Alphabet code for type of programme
01	Programme code
C	Type of course
1	Semester I – semester number
01	First course of semester – course serial number It will be XX for the elective/choice based courses

Semester VII

(For Students NOT OPTING Semester Long Internship

&

For Students OPTING FOR SEMESTER LONG INTERNSHIP In semester VIII)

Credit Scheme

Course Code	Course Name	Teaching Scheme (Hrs.) Per Week TH – P – TUT	Total (Hrs.) Per week	Credits Assigned TH – P – TUT	Total Credits	Course Category
216U01C701	Software Design Thinking	3 – 2 – 0	5	3 – 1 – 0	4	PC
216U01E73X	Departmental Elective-III	3 – 2 – 0	5	3 – 1 – 0	4	PE
216U01E74X	Departmental Elective-IV	3 – 2 – 0	5	3 – 1 – 0	4	PE
216U06O7XX	Open Elective Technical (College offered/ NPTEL/ Coursera)	3 – 0 – 0	3	3 – 0 – 0	3	OE
216U01P701	Project- II*	0 – 10 – 0	10	0– 5 – 0	5	PR
	Total	**12– 16 – 0**	**28**	**12 – 8 – 0**	**20**	

***Project-II is a continuation of the Project-I course undertaken in Semester VI.**

Examination Scheme

Course Code	Course Name	Examination Scheme				
		Marks				
		CA		ESE	Lab / Tut CA	
		ISE	IA			Total
216U01C701	Software Design Thinking	30	20	50	50	**150**
216U01E73X	Departmental Elective-III	30	20	50	50	**150**
216U01E74X	Departmental Elective-IV	30	20	50	50	**150**
216U06O7XX	Open Elective Technical (College offered/ NPTEL/ Coursera)	30	20	-	-	**50**
216U01P701	Project- II	-	-	100#	50$	**150**
	Total	**120**	**80**	**250**	**200**	**650**

$Continuous assessment of the project shall be carried out by an internal panel throughout the semester. The evaluation will be based on the quality and progress of the project work, as well as periodic oral presentations and demonstrations.# Evaluation at the end of the semester shall be conducted jointly by an external examiner (preferably from industry) and an internal examiner. The total marks shall be 100, of which 70 marks will be awarded based on the quality of the project work, final project report, oral presentation, and demonstration. The remaining 30 marks will be allotted based on the significant outcomes of the project, such as participation in project competitions, publication of research papers, project deployment, copyrights, patents, or other achievements, as assessed by the examiners.

Semester VIII

Credit and Examination Scheme

Semester VIII

Teaching and Credit Scheme

(For Students NOT OPTING Semester Long Internship)

Course Code	Course Name	Teaching Scheme (Hrs.) Per Week TH – P – TUT	Total (Hrs.) Per week	Credits Assigned TH – P – TUT	Total Credits	Course Category
216U01E85X	Departmental Elective-V	2 – 2 – 0	4	2 – 1 – 0	3	PE
216U01E86X	Departmental Elective-VI	2 – 2 – 0	4	2 – 1 – 0	3	PE
216U01P801	Project III/ Research project*	0 – 14 – 0	14	0 – 7 – 0	7	PR
	Total	**4 – 18 – 0**	**22**	**4 – 9 – 0**	**13**	

***Project-III is a continuation of the Project-II course undertaken in Semester VII OR it may be offered as standalone project, subject to approval by the internal evaluation panel**

For Students OPTING FOR SEMESTER LONG INTERNSHIP In semester VII

Course Code	Course Name	Total Credits	Course Category
216U01P801	Semester long Internship$	10	PR
216U01E85x	Online NPTEL / Coursera course*	3	PE
	Total	**13**	

Examination Scheme

(For Students NOT OPTING Semester Long Internship)

Course Code	Course Name	Examination Scheme				
		Marks				
		CA		ESE	Lab / Tut CA	
		ISE	IA			Total
216U01E85X	Departmental Elective-V	30	20	50	50	**150**
216U01E86X	Departmental Elective-VI	30	20	50	50	**150**
216U01P801	Project III/ Research project	-	-	100*	100$	**200**
	Total	**60**	**40**	**200**	**200**	**500**

$ Continuous assessment of the project shall be carried out by an internal panel throughout the semester. The evaluation will be based on the quality and progress of the project work, as well as periodic oral presentations and demonstrations.

Evaluation at the end of the semester shall be conducted jointly by an external examiner (preferably from industry) and an internal examiner. The total marks shall be 100, of which 70 marks will be awarded based on the quality of the project work, final project report, oral presentation, and demonstration. The remaining 30 marks will be allotted based on the significant outcomes of the project, such as participation in project competitions, publication of research papers, project deployment, copyrights, patents, or other achievements, as assessed by the examiners.

For Students OPTING FOR SEMESTER LONG INTERNSHIP In semester VII

Course Code	Course Name	Examination Scheme				
		Marks				
		CA		ESE	Lab / Tut CA	
		ISE	IA			Total
216U01P801	Semester long Internship$	@100	-	@200	-	**300**
216U01E85x	Online NPTEL / Coursera course*	-	-	-	-	**-**
	Total	**100**	**-**	**200**	**-**	**300**

@ - To be conducted by Pair of Internal Examiners approved by department. 150 marks for presentation/examination conducted by internal examiner and Industry expert approved by department and 50 marks will be given on Industry Feedback evaluation

For details of Semester long internships, refer to Internship cell policy and Documents.

Note:

As per college internship policy, it is mandatory for every student to complete 10 weeks of Internship spanning over the four years of B. Tech Programme which is over and above the academic credits.

Students can take up internships in community services / socially relevant projects (optional and limited to 4 weeks) and in the technical domain (minimum 6 weeks or more).

Students will be awarded an internship completion certificate along with their graduation.

Departmental Elective - III for VII Semester

Sr. No.	Course Code	Name of Elective	Remarks
1	216U01E731	Advanced Cloud Computing	With pre-requisite of previous CC course
2	216U01E732	Big Data Analytics	May not available to students with Honours in Data Science and Analytics
3	216U01E733	Computer Simulation and Modelling	
4	216U01E734	Cyber Security, Forensics and Cyber Law	May not available to students with Honours in Cyber Security and Forensics
5	216U01E735	Geographic Information System and Spatial Computing	May not available to students with Honours in Geospatial Computing
6	216U01E736	Reinforcement Learning	
7	216U01E737	Edge AI	

Departmental Elective - IV for VII Semester

Sr. No.	Course Code	Name of Elective	Remarks
1	216U01E741	Block Chain Technology	
2	216U01E742	Computer Vision	
3	216U01E743	Advanced Web Technology	
4	216U01E744	DevOps	
5	216U01E745	Natural Language Processing	
6	216U01E746	Secure Coding	May not available to students with Honours in Cyber Security and Forensics
7	216U01E747	Software Architecture	

Departmental Elective – V for VIII Semester

Sr. No.	Course Code	Name of Electives	Remarks
1	216U01E751	Advanced Algorithms: Design and Analysis	
2	216U01E752	Bioinformatics	
3	216U01E753	Business Analytics	May not available to students with Honours in Data Science and Analytics
4	216U01E754	Cyber Physical Systems	
5	216U01E755	Deep Learning	
6	216U01E756	Internet of Everything	

Departmental Elective – VI for VIII Semester

Sr. No.	Course Code	Name of Electives	Remarks
1	216U01E761	Blockchain Technology : platforms and applications	
2	216U01E762	Game Programming	
3	216U01E763	High Performance Computing	
4	216U01E764	IoT Security	With pre-requisite of previous IoT course
5	216U01E765	NLP and introduction to LLM	
6	216U01E766	MLOps	With pre-requisite of previous DevOps course

Open Electives Technical 2 for VII/VIII Semester

Sr. No.	Course Code	Name of the Elective
1	216U01O701	Genetics Algorithm and Applications
2	216U01O702	Reinforcement Learning
3	216U01O703	Web Analytics and Web Intelligence
4	216U01O704	Massive Graph Analysis
5	216U01O705	Audio Signal Processing
6	216U01O706	Applied Machine Learning using Tensor flow

	Semester VII		Semester VIII	
Thrust Area	**Elective III**	**Elective IV**	**Elective V**	**Elective VI**
Intelligent System and Data Processing	Big Data Analytics	Natural Language Processing	Bioinformatics	Game Programming
	Reinforcement Learning		Business Analytics	Natural Language Processing with Deep Learning
Network System and security	Cyber Security, Forensics and Cyber Law	Block Chain Technology	Cyber Physical Systems	IoT Security
	Adv. Cloud Computing	Secure Coding		Blockchain Technology : platforms and applications
Image analysis and interpretation	Geographic Information System and Spatial Computing	Computer Vision	Deep Learning	
System and software engineering	User Experience Design	Advanced Web Technology	Advanced Algorithms: Design and Analysis	High Performance Computing
	Computer Simulation and Modelling	Software Architecture	Internet of Everything	MLOps

Semester VII

Course Code	Name of the Course			
216U01C701	Software Design Thinking			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course prerequisites:

Project-Based Learning, and Software Engineering

Course Objectives:

1. Develop a structured problem-solving mindset grounded in design thinking.
2. Enable students to generate, evaluate, and select effective design alternatives.
3. Encourage integration of creativity, technology, and practicality in software solution design.
4. Foster the ability to design sustainable, real-world software solutions addressing user needs and societal challenges.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO 1	Understand the principles, process, and importance of design thinking in software development.
CO 2	Apply design thinking methodology to analyze user needs and define problem statements.
CO 3	Generate innovative design ideas and solutions using brainstorming, prototyping, and ideation techniques.
CO 4	Evaluate and refine software design alternatives through user feedback, usability testing, and iteration.
CO 5	Develop a complete software design prototype addressing real-world engineering problems with creativity and feasibility

Module No.	Unit No.	Contents	No of Hrs.	CO

1	**Introduction to Software Design Thinking**		**09**	CO 1 CO 2
	1.1	Concept of design thinking in software engineering		
	1.2	Stages: Empathize, Define, Ideate, Prototype, Test		
	1.3	Human-centered software design approach		
	1.4	Design vs. traditional problem-solving		
	1.5	Case studies on successful design thinking in software		
2	**Empathize & Define**		**09**	CO 1 CO 2
	2.1	Understanding user needs, pain points, and requirements		
	2.2	Conducting user research: interviews, surveys, observation		
	2.3	Creating empathy maps and personas		
	2.4	Problem framing and defining point-of-view (POV) statements		
	2.5	Translating requirements into design goals		
3	**Ideation in Software Design**		**09**	CO 2 CO 3 CO 4
	3.1	Creativity techniques: brainstorming, SCAMPER, mind mapping		
	3.2	Generating multiple design alternatives		
	3.3	Storyboarding and user journey mapping		
	3.4	Design heuristics for software solutions		
	3.5	Evaluating idea feasibility and prioritization		
4	**Prototyping Techniques**		**09**	CO 3 CO 4 CO 5
	4.1	Types of prototypes: low-fidelity vs. high-fidelity		
	4.2	Tools for software prototyping (wireframes, mockups, clickable prototypes)		
	4.3	Agile and iterative prototyping		
	4.4	Interaction design and UI/UX considerations		
	4.5	Rapid prototyping case study in software development		
5	**Testing, Feedback & Implementation**		**09**	CO 3 CO 4 CO 5
	5.1	Usability testing principles		
	5.2	Methods of collecting user feedback		
	5.3	Iteration cycles and refinement strategies		
	5.4	Translating prototypes into software architecture/design models		
	5.5	Capstone project: applying design thinking to solve a real-world software problem		
		Total	**45**	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Tim Brown	*Change by Design: How Design Thinking Creates New Alternatives for Business and Society*	Harper Business	Kindle Edition 2019
2	Nigel Cross	*Design Thinking: Understanding How Designers Think and Work*,	Bloomsbury	2nd Edition 2023
3	Jeanne Liedtka & Tim Ogilvie	*Designing for Growth: A Design Thinking Tool Kit for Managers*	Columbia Business School Publication	2nd edition 2019
4	Pahl, Beitz, Feldhusen, Grote	*Engineering Design: A Systematic Approach*	Springer	3rd edition 2009
5	Kees Dorst	*Frame Innovation: Create New Thinking by Design*	MIT Press.	Illustrated edition 2015
6	https://www.interaction-design.org/literature/topics/design-thinking			Last accessed on Sept 3, 2025
7	https://www.pluralsight.com/courses/design-thinking-for-engineers			Last accessed on Sept 3, 2025
8	https://www.nngroup.com/articles/design-thinking/			Last accessed on Sept 3, 2025
9	https://dschool.stanford.edu/resources			Last accessed on Sept 3, 2025
10	https://mitsloan.mit.edu/ideas-made-to-matter/design-thinking-explained			Last accessed on Sept 3, 2025

Laboratory/Tutorial

Course Code	Name of the Course			
216U01L701	Software Design Thinking-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Departmental Elective III

Course Code	Name of the Course			
216U01E731	Advanced Cloud Computing			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Fundamental knowledge on Operating system.
- Basics of client/server programming and network protocols.
- Basics of cloud computing

Course Objectives:

1. To introduce basic computing models and functions of those models.
2. To understand the current practices in cloud computing.
3. To understand the distributed computing models and technologies.
4. To understand the other advanced and research topics in cloud computing.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Describe basics of cloud computing and its fundamentals.
CO2. Evaluate and apply concepts of SLA management and cloud economics to ensure service quality in cloud delivery models.
CO3. Analyze and apply relational, cloud, and parallel database architectures to design efficient and scalable data management solutions.
CO4. Leverage cloud-based ML/AI techniques for scalable learning and predictive analytics.
CO5. Apply Sensor cloud, IoT, and fog computing for smart services.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Cloud Computing**		6	CO1
	1.1	Introduction to cloud computing, Computing platforms and technologies, Types of clouds. Cloud service models, Cloud applications,Virtualization Concepts, Types of Virtualization.		
	1.2	#Self-learning: Hardware Support for Processor, Memory, I/O Virtualization at Intel		
2	**Service Management in Cloud Computing**		10	CO2
	2.1	SERVICE LEVEL AGREEMENT (SLA), SLA Contents, Types of SLA, Service Level Objectives (SLOs), SLA Requirements, Key Performance Indicators (KPIs), Metrics for Monitoring and Auditing, SLA Requirements w.r.t. Cloud Delivery Models, Examples of Cloud SLAs, Limitations of SLA, Numericals on Service Availability, SLA Violation and Penalty Computation		
	2.2	Cloud Economics: Economic Viewpoint, Utility Pricing in Detail, Value of on-Demand Services, Penalty Costs for Exponential Demand		
3	**Managing Data**		10	CO3
	3.1	Relational databases:Data Storage Techniques, Default data storage and retrieval mechanism since 80s		
	3.2	Cloud File Systems: Google File System (GFS Architecture), BigTable Storage, Dynamo Architecture, Datastore		
	3.3	Parallel Database Architectures, MapReduce Model: Fault Tolerance Efficiency, Applications		
4	**Cloud Computing and ML/AI**		09	CO4
	4.1	Large-scale machine learning service on clouds, Resource management for production-scale machine learning.		
	4.2	Applied machine learning research in clouds. i.e., workload and error prediction, Current and future (system specific) research challenges for AI and ML		
5	**Sensor Clouds, Cloud IoT and Fog Computing**		10	CO5
	5.1	Sensor Clouds,Overview of Sensor Cloud Infrastructure, Virtual Sensors, System Architecture of Sensor Cloud Infrastructure, Virtual Sensor Configurations-one-to-many, many-to-one, and many-to-many, and derived		
	5.2	Basic IoT Architecture, Cloud Components for IoT, iCOMOT: An IoT Cloud System, Infrastructure, Protocols and Software Platforms for establishing an Internet of Things (IoT) Cloud system, Architecture for IoT-based vehicular data clouds, Architecture for Intelligent Parking Cloud service		
	5.3	Fog Computing, Cloud vs Fog Computing, Cloud-Fog-Edge-Computing, Case Study-Health Cloud Fog Model		
	5.4	**Self-Learning: Edge Computing Architecture**		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Rajkumar Buyya, Christian Vecchiola, S Thamarai Selvi	Mastering Cloud Computing	McGraw Hill Education Private Limited	2013
2	Judith Hurwitz, R.Bloor,M.Kanfman, F.Halper	OpenStack Cloud Computing Cookbook	PACKT Publishing BIRMINGHAM - MUMBAI	Third Edition
3	J.Vette, Toby J. Vette, Robert Elsenpeter	Cloud Computing: A Practical Approach	Tata McGraw Hill	1st, 2009
4	Rajkumar Buyya, James Broberg, Andrzej Goscinski	Cloud Computing, Principles and Paradigms	Wiley	1st ,2013

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL731	Advanced Cloud Computing - Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E732	Big Data Analytics			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Database management system, Data mining

Course Objectives:

1. To provide an overview of the growing field of Big Data Analytics and its role in solving real-world challenges across domains
2. To introduce tools and platforms such as Hadoop, Spark, NoSQL, and MapReduce required to manage and analyze large-scale data..
3. To teach the fundamental techniques and principles in achieving scalable and streaming analytics.
4. To enable students to apply Big Data Analytics for complex real-world decision-making problems in areas such as business, healthcare, and sustainability

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1: Interpret the fundamental enabling techniques like Hadoop, MapReduce, Spark in achieving Big data analytics.

CO2: Apply techniques for finding similarity, distance measures, and dimensionality reduction in large datasets.

CO3: Analyze business models, scientific computing paradigms, and scalable algorithms to solve real-world big data problems.

CO4: Demonstrate Big Data Analytics applications in domains such as recommender systems, healthcare, social networks etc

Module No.	Unit No.	Contents	No of Hrs.	CO
1		**Introduction to Big Data and Framework**	10	CO1
	1.1	Introduction to Big Data, Big Data characteristics, types of Big Data, Traditional vs. Big Data business approach, Case Study of Big Data Solutions.		
	1.2	Introduction to Hadoop and its components		
	1.3	**Distributed File Systems:** Physical Organization of Compute Nodes,Large-Scale File-System Organization **MapReduce:** The Map Tasks, Grouping by Key, The Reduce Tasks, Combiners, Details of MapReduce Execution, Coping With Node Failures.		
	1.4	**Algorithms Using MapReduce:** Matrix-Vector Multiplication by MapReduce, Relational-Algebra Operations, Computing Selections by MapReduce, Computing Projections by MapReduce, Union, Intersection, and Difference by MapReduce, Computing Natural Join by MapReduce, Grouping and Aggregation by MapReduce, Matrix Multiplication, Matrix Multiplication with One MapReduce Step.		
	1.5	Introduction to Spark: architecture (RDD, DataFrames, Datasets), Spark Core, Spark SQL, Spark Streaming, MLlib, GraphX Spark Programming Model: Transformations, Actions, Lazy evaluation, DAG execution Comparison of Hadoop MapReduce vs. Spark (batch vs. in-memory, performance, use cases)		
		Self-Learning: Cloud-based big data platforms (AWS EMR, Google BigQuery, Azure HDInsight)		
2		**Finding Similar Items and Dimensionality Reduction**	08	CO2
	2.1	Applications of Near-Neighbor Search- Jaccard Similarity of Sets, Similarity of Documents, Collaborative Filtering as a Similar-Sets Problem.		
	2.2	Distance Measures: Definition of a Distance Measure, Euclidean Distances, Jaccard Distance, Cosine Distance, Edit Distance, Hamming Distance.		
	2.3	Shingling of Documents, Similarity-Preserving Summaries of Sets, Locality-Sensitive Hashing for Documents, Applications of Locality-Sensitive Hashing		
	2.4	Dimensionality Reduction: Eigenvalues and Eigenvectors of Symmetric Matrices, Principal-Component Analysis, Singular-Value Decomposition CURE Decomposition		
3		**Mining Data Streams**	09	CO3
	3.1	The Stream Data Model: A Data-Stream-Management System, Examples of Stream Sources (sensor networks, healthcare		

		wearables, social media, smart city IoT) , Stream Query, Issues in Stream Processing.		
	3.2	Sampling Data in a Stream: Obtaining a Representative Sample, The General Sampling Problem, Varying the Sample Size.		
	3.3	Filtering Streams: The Bloom Filter, Analysis		
	3.4	Counting Distinct Elements in a Stream: The Count-Distinct Problem, The Flajolet-Martin Algorithm, Combining Estimates, Space Requirements. Estimating Moments Counting Ones in a Window: The Cost of Exact Counts, The Datar-Gionis-Indyk-Motwani Algorithm. Real-time stream analytics in Healthcare (remote patient monitoring, epidemic tracking)		
		#Self-Learning: Kafka, Spark Streaming		
4	**Link Analysis And Frequent Itemsets**		09	CO3 CO4
	4.1	PageRank Definition, Structure of the web, dead ends, Using Page rank in a search engine, Efficient computation of Page Rank: PageRank Iteration Using MapReduce, Use of Combiners to Consolidate the Result Vector. Topic sensitive Page Rank,Hubs and Authorities		
	4.2	Handling Larger Datasets in Main Memory Algorithm of Park, Chen, and Yu, All or Most frequent itemsets in two passes.		
	4.3	The SON Algorithm and MapReduce, Toivonen's Algorithm		
5	**Clustering and Mining Social Network and graphs**		09	CO3 CO4
	5.1	CURE Algorithm, Clustering in Non-Euclidean Spaces Stream-Computing , A Stream-Clustering Algorithm, Initializing & Merging Buckets, Answering Queries		
	5.2	Recommendation Systems – A model for Recommendation systems, Content based recommendation, Collaborative Filtering		
	5.3	Mining Social-Network Graphs – Social networks as graphs, Clustering, Direct discovery of communities, Partitioning of Graphs, Finding overlapping of communities, SimRank, Counting Triangles Case Study: Mining Urban Sustainability Graphs		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Anand Rajaraman and Jeff Ullman	Mining of Massive Datasets	Cambridge University Press	Third Edition , 2020
2	Alex Holmes	Hadoop in Practice	Manning Press, Dreamtech Press.	Second Edition , 2015
3	Bill Franks	Taming The Big Data Tidal Wave: Finding Opportunities In Huge Data Streams With Advanced Analytics	Wiley	2012
4	Mayuri Mehta , Kalpdrum Passi, Indranath Chatterjeee, Rajan Patel	Knowledge Modelling and Big Data Analytics in Healthcare Advances and Applications	CRC Press	2022
5	Radha Shankarmani M. Vijaylakshmi	Big Data Analytics	Wiley	2st edition, 2018
6	Perrin, Jean-Georges	Spark in Action	O'Reilly Media Inc.	2020
7	https://web.stanford.edu/class/cs246/			
8	https://onlinecourses.nptel.ac.in/noc20_cs92/preview Coursera - https://www.coursera.org/learn/scala-spark-big-data			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL732	Big Data Analytics - Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E733	Computer Simulation and Modeling			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Understanding of basic concepts of probability theory, algorithms and data structures.

Course Objectives:

1. Introduction to discrete event simulation systems
2. Modeling techniques of entities, queues, resource
3. Formulate and build valid models, implement the model
4. Perform simulation analysis of the system and analyze results

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand the fundamentals of **simulation** and apply **manual simulation** techniques to queuing and **inventory management** problems.

CO2. Model and analyze queuing systems, including M/M/1, M/G/1, multi-class queuing systems and apply these models to optimize performance in real-world systems.

CO3. Generate pseudorandom numbers and perform statistical tests to measure the quality of a pseudorandom number generator.

CO4. Analyze the systems for input modeling and output performance.

CO5. Understand and apply advanced simulation techniques to model complex systems in simulation applications.

Module No.	Unit No.	Contents	No of Hrs.	CO
1		**Introduction to Simulation, Simulation Examples & general principles**	12	CO1
	1.1	Introduction to Simulation: advantages, disadvantages, types of models & steps in simulation study.		
	1.2	Simulation Examples: Manual Simulation: Queuing and Inventory problems		
	1.3	General Principles: Event Scheduling Algorithm/Time advance algorithm, Simulation using time advance algorithm.		
2		**Statistical & Queuing Models**	08	CO2
	2.1	Queuing Models: M/G/1, M/M/1		
	2.2	Multi-Class Queuing Systems: Modeling systems with multiple classes of customers or services (e.g., M/M/c, M/G/c).		
	2.3	Time-Dependent Queuing Models: Non-stationary processes, such as Non-Homogeneous Poisson Processes (NHPP)		
	2.4	#Self-Learning: Statistical Models in simulation: Discrete and Continuous Distributions & M/M/1 with priority classes.		
3		**Random Number Generation**	10	CO3
	3.1	Random Number Generation Techniques		
	3.2	Testing random numbers: Chi square, K-S, Runs up and Down test, Gap, Poker and autocorrelation tests.		
	3.3	Random Variate Generation: Inverse transform technique, Convolution Method & Acceptance-Rejection Technique		
		#Self-Learning: Quasi-Monte Carlo for random number generation		
4		**Analysis of Simulation Data**	08	CO4
	4.1	**Input Modeling** : Data collection, Identifying the Distribution with Data, Parameter Estimation, Goodness-of-Fit Tests,		
	4.2	Selecting Input Models without Data, Multivariate and Time-Series Input Models		
	4.3	**Output Analysis:** Estimation of absolute performance, Output Analysis Concepts, output analysis for steady state simulation.		
5		**Advanced Simulation Techniques**	07	CO5
	5.1	Overview of Simulation Methods: Introduction to Agent-Based Modeling (ABM), System Dynamics Modeling (SDM).		
	5.2	Agent-Based Modeling: Understanding agents, environment, and interactions. Components: Agents, behaviour, environment, rules. Applications of Agent-Based Modeling.		
	5.3	System Dynamics Modeling: Key Concepts of SDM: Stocks and Flows: Core elements of SDM models. Feedback Loops and Delays. Applications of System Dynamics Modeling.		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1.	Jerry Banks, John Carson, Barry Nelson, and David M. Nicol	Discrete Event System Simulation	Prentice-Hall	Fifth Edition
2.	Averill M Law	System Modeling & Analysis	Tata McGraw Hill	Fourth Edition
3.	Lawrence M. Leemis Stephen K. Park	Discrete-Event Simulation: A First Course	Pearson	First Edition
4.	Banks C M , Sokolowski J A	Principles of Modeling and Simulation	Wiley	2010
5.	Uri Wilensky, William Rand	An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo	MIT Press	2015

Laboratory/Tutorial

Course Code	Computer Simulation and Modelling			
216U01EL733	Computer Simulation and Modeling- Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E734	Cyber Security, Forensics & Cyber Law			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Computer Organization & Architecture, System Security, Computer Networks.

Course Objectives:

1. Understand basics of information security, crimes and defence.
2. Learn principles and process of digital forensics.
3. Analyse and interpret disk forensic evidence.
4. Use tools to extract and investigate network evidence.
5. Know legal aspects of cyber-crimes under IT Act.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Identify various security goals, computer crimes & methods of defence.
CO2. Understand the fundamentals of digital forensics.
CO3. Analyze and interpret the results of disk forensic operations
CO4. Apply forensic tools to extract and investigate the evidence from network.
CO5. Relate the corresponding computer security acts with the crimes.

Module No.	Unit No.	Contents	No of Hrs.	CO
1		**Introduction to Cybersecurity**	05	CO1
	1.1	Definition, need, and scope of cybersecurity, Cybercrimes: origins and evolution, Classification of cybercrimes (financial, data theft, cyber terrorism, hacktivism, cyberbullying, etc.)		
	1.2	Cyberspace and its global dimensions, Cyber profiling (methods, digital footprint analysis), Role of AI/ML in modern cybersecurity (threat detection and anomaly profiling)		
	1.3	Indian cybersecurity initiatives (CERT-In, Cyber Dost, National Cybersecurity Strategy)		
2		**Data Privacy and Theft**	10	CO2
	2.1	**Data Theft and Attacks:** Adwares, malwares, ransomwares, trojans, spywares, keyloggers, rootkits, Phishing (types: spear phishing, whaling, clone phishing), SQL injection attacks and prevention, Social engineering attacks (vishing, smishing, Juice jacking) Data Theft Preventions (Hardening Techniques – Server, Database, OS, Network)		
	2.2	**Identity Theft:** Types of identity theft: financial, medical, criminal, synthetic identity theft, Deepfakes. Prevention techniques: MFA, biometrics, digital hygiene practices. Case study on data breaches (Aadhaar, Zomato etc).		
	2.3	Issues surrounding data privacy, Data privacy vs. Data security, Data privacy mechanisms (anonymization, pseudonymization, consent-based sharing), Legislations on data privacy – local and global, Guidelines for data privacy (OECD principles, Digital Personal Data Protection Act 2023, GDPR, HIPAA), CCPA (California Consumer Privacy Act)		
	2.4	#Self-Learning: Case study on identity theft, Data theft and Privacy.		
3		**Digital Forensics Fundamentals**	10	CO3
	3.1	**Introduction to Digital Forensics** Six A's of digital forensics (Awareness, Acquisition, Authentication, Analysis, Assessment, Action), Digital evidence and its characteristics, Sources of Digital evidence (Logs, Network, Databases, End Point devices etc), Phases of Digital investigations, Incident response and its methodology		
	3.2	**Classification and Handling of Digital Evidence** Volatile vs. non-volatile evidence, Rules and guidelines for extraction of digital evidence, Forensic duplicates and imaging, Establishing chain of custody, Admissibility of evidence in the		

		court of law, Role of hash functions (MD5, SHA-256) in verifying evidence integrity		
	3.3	**Information Retrieval and Recovery** Information retrieval and recovery techniques, Cloning and imaging techniques, Password cracking methods and ethical considerations, Data carving and recovery from file systems and mobile devices, Forensics audit and reporting, Tools for forensic investigation (EnCase, Autopsy, FTK Imager, Wireshark, Volatility framework – awareness level), Anti-forensics and counter-forensics methods with real-world examples		
	3.4	#Self-Learning: Introduction to cloud forensics (unique challenges in cloud-based investigations)		
4	**Network Forensics**		10	CO4
	4.1	**Network-Based Attacks** MITM, OWASP Top 10 vulnerabilities, ARP spoofing, IP & MAC spoofing, DNS attacks (DNS cache poisoning, tunnelling), SYN flooding attacks, port scanning, DoS & DDoS attacks		
	4.2	**Sources of Network Evidence** Digital footprints, Email logs and attachments, Web usage records (cookies, browser cache, history), Network traffic logs (packet captures, NetFlow records), Email forensics & investigations (phishing header analysis, trace routes), Cloud network evidence (logs from AWS CloudTrail, Azure Monitor, etc.), Mobile app traffic evidence (WhatsApp/Facebook packet captures with Wireshark demo)		
	4.3	**Network Forensics and Tools** Browser forensics, Email Forensics, Network Forensics: Network Analysis and Assessment using Nmap, Nessus, Wireshark, NetworkMiner etc. Kali Linux tools, Metasploit, Metasploitable, Deep Web and Dark Web exploration, Network Log Analysis using Security Information and Event Management (SIEM) basics (e.g., Wazuh, Splunk, ELK stack awareness), PCAP analysis Wireshark/TCPDump		
5	**Cyber Laws and Acts**		10	CO5
	5.1	**Introduction to Cyber Law:** Cyber ethics, Software Piracy, Intellectual Property, IP Theft, Copyright, Trademark, Privacy and Censorship. Digital Personal Data Protection Act (DPDP) 2023, Intermediary Guidelines and Digital Media Ethics Code 2021 Role of Budapest Convention on Cybercrime		
	5.2	**Intellectual Property Issues in Cyberspace:** Computer software & copyright law. Software licenses (GPL, Apache,		

		MIT, Creative Commons), Domain names & cybersquatting disputes (ICANN, WIPO), Trademark issues in cyberspace,		
	5.3	**Self-reading:** Open-source licensing challenges in AI/ML tools, Case law discussion on copyright of digital content (e.g., music/OTT platforms)		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Bill Nelson, Amelia Phillips, Christopher Steuart.	Guide to Computer Forensics and Investigations.	Cengage Learning, USA	6th Edition paperback, 2022
2	Jason T. Luttgens, Mathew Pepe, Kevin Mandia	Incident Response and Computer Forensics.	Tata McGraw Hill Education	3rd Edition, 2014.
3	Marie-Helen Maras	Computer Forensics: Cybercriminals, Laws and Evidences	Jones and Bartlett Learning	2nd Edition, 2014
4	Davidoff Ham	Network Forensics Tracking Hackers through Cyberspace	Pearson India	1st Edition, 2013.
5	Adv. Prashant Mali	Cyber Law and Cyber Crimes Simplified	Cyber Infomedia	January 2017.
6	https://lawbhoomi.com/cyber-law-notes/			
7	https://onlinecourses.swayam2.ac.in/cec21_ge10/preview https://onlinecourses.nptel.ac.in/noc23_cs127/preview			

Laboratory/Tutorial

Course Code	Computer Simulation and Modelling			
216U01EL734	Cyber Security, Forensics & Cyber Law-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E735	GIS & Spatial Computing (GISSC)			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basic Programming, Database Fundamental, Mathematics and Statistics.

Course Objectives:

1. Provide foundational knowledge of Geographic Information Systems (GIS) concepts, components, and data models
2. Develop the ability to process, analyze, and interpret spatial data using vector and raster models
3. Equip students with skills for spatial analysis, modeling, and decision-making in real-world applications
4. Introduce modern GIS tools, technologies, and integration with big data and machine learning frameworks.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand of core concept that define the Geographical Information System
CO2. Apply vector and raster data analysis techniques for spatial problem-solving.
CO3. Perform surface mapping, interpolation, and network analysis for spatial decision-making.
CO4. Develop and interpret spatial models using regression and spatial dependency tests.
CO5. Implement spatial data mining techniques and evaluate advanced GIS applications.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to GIS and Data Models**		10	CO1
	1.1	Components of GIS, Brief history of GIS, Elements of GIS, Application of GIS.		
	1.2	**Spatial data**: Vector space model: Representation of spatial features, Topology, Georelational Data model, Object-Based data model, Representation of complex features		
	1.3	Raster Data Model: Elements of Raster data model, Satellite images, Digital elevation models, Other types of Raster Data, Raster data structure, Raster data compression, Data conversion and Integration.		
		Self-learning: Spatial Referencing: Reference surfaces for mapping, Coordinate Systems, Map Projections, Coordinate Transformations		
2	**Core Data analysis**		08	CO2
	2.1	Buffering, Overlay, Distance measurement , Pattern analysis, Feature Manipulation		
	2.2	Local operation, Neighborhood Operations, Zonal operations, Physical distance Measure Operations, other Raster data operations, Map algebra, Comparison of Vector and Raster data analysis		
		#Self-Learning: Error Propagation in spatial data processing: How Errors propagate, Quantifying error propagation		
3	**Surface Mapping and Analysis**		09	CO3
	3.1	Data for Terrain Mapping and analysis, Terrain Mapping, slope and aspect, Surface curvature, Raster vs TIN , Viewshed and water shed analysis .		
	3.2	Elements of Spatial Interpolation, Global Methods, Local Methods, Kriging, Comparison of spatial interpolation		
	3.3	Least-cost path analysis and its application, Network analysis, Geocoding and segmentation		
		#Self-Learning: Stages of spatial data handling,QGIS, Hadoop, GeoSpark R PostgreSQL,		
4	**Spatial Modeling**		09	CO4
	4.1	Descriptive models, Explanatory models		
	4.2	Modeling area data: Spatial Regression Models, Tests for Spatial Dependence		
	4.3	The Spatial Durbin Model , Estimation of Spatial Regression Models, Model Parameter Interpretation		

		Self-reading: advance algorithms in GIS,QGIS, Hadoop, GeoSpark R PostgreSQL		
5	GIS and Spatial Analysis		08	CO5
	5.1	Spatial Association rule mining, spatial hierarchal clustering, Set based decision and knowledge recovery		
	5.2	Case study: Advanced GIS application for Spatial data mining .		
		Self-reading: GIS and spatial data analysis using Deep and ML algorithms		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Kang-Tsung, Chan	Introduction to Geographical Information System	McGraw Hill Education	Ninth Edition, 2018
2	Manfred M. Fischer and Jinfeng Wang	Spatial Data Analysis: Models, Methods and Techniques	Springer Briefs ISBN: 978-3-642-21719-7	January 2011
3	Robert Haining, Robert Haining	Spatial Data Analysis Theory and Practice	Paperback Cambridge University Press	17 April 2003
4	Deren Li, Shuliang Wang, Deyi Li	Spatial Data Mining Theory and Application	Springer ,Science Press	2013
5	Michael J De Smith (Author), Michael F Goodchild (Author), Paul a Longley	Geospatial Analysis: A Comprehensive Guide	Winchelsea Press	Sixth Edition 2018

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL735	GIS & Spatial Computing (GISSC)-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E736	Reinforcement Learning			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Mathematics and Statistics, Machine Learning Fundamentals, Programming Skills

Course Objectives:

1. Comprehension of fundamentals of reinforcement learning
2. Application of various RL methods for
3. Analysis using temporal difference and deep learning methods
4. Evaluation of various methods of RL for application development

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Interpret fundamentals of Reinforcement learning methods
CO2. Apply various dynamic programming and Monte-Carlo methods
CO3. Apply different temporal difference learning policies
CO4. Use n-step bootstrapping policies and apply planning for different models
CO5. Understand future of reinforcement learning

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Reinforcement Learning**		08	CO1
	1.1	Elements of Reinforcement Learning, Limitations and scope		
	1.2	Markov Decision Processes to solve real-world problems, Goals and rewards, Policies and value functions		
	1.3	Bellman equations for Markov Decision Processes		
		Self-learning: History of RL		
2	**Dynamic Programming and Monte Carlo Methods**		10	CO2
	2.1	Iterative policy evaluation, Policy improvement, Policy iteration and value iteration		
	2.2	Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency of Dynamic Programming		
	2.3	Classic Monte Carlo prediction and control methods, Greedy and epsilon-greedy policies, Exploration-Exploitation Dilemma and its solutions		
	2.4	#Self-Learning: A/B testing and bandit algorithm		
3	**Temporal Difference Learning**		10	CO3
	3.1	TD Prediction ,Advantages of TD Prediction Methods		
	3.2	Sarsa: On-policy TD Control		
	3.3	Q-learning: Off-policy TD Control		
	3.4	#Self-Learning:Deep learning, Generative models		
4	**n-step Bootstrapping**		09	CO4
	4.1	n-step TD Prediction, n-step Sarsa, n-step Off policy		
	4.2	Off policy Learning Without Importance Sampling: The n-step Tree Backup Algorithm		
	4.3	Models and Planning, Dyna: Integrated Planning, Acting, and Learning		
	4.4	#Self-reading: n-step TD learning eligibility trace proofs		
5	Deep Q-Learning and Reinforcement Learning Lifecycle		08	CO5
	5.1	Deep Q-learning: Experience Replay, Implementing DQ N with examples.		
	5.2	RL project life cycle, problem definition for RL projects		
	5.3	Future of reinforcement learning, RL market opportunities, Future RL and research directions		
	5.4	#Self-reading: RL debugging		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Richard S. Sutton and Andrew G. Barto	*Reinforcement Learning: An Introduction*	The MIT Press	Second Edition, 2018
2	Stuart Russell and Peter Norvig	*Artificial Intelligence: A Modern Approach*	Pearson	3 rd edition, 2010
3	Robert Tibshirani, and Jerome Friedman	*The Elements of Statistical Learning, Trevor Hastie*	Springer	2nd edition, 2009
4	M. Gopal	*Applied Machine Learning*	Mc-Graw Hill Education India Pvt. Ltd.	Print edition: ISBN-13: 978-93-5316-025-8,
5	Chris Bishop	*Pattern recognition and machine learning (PRML)*	Springer	ISBN-13: 978-0387-31073-2, 2006
6	Phil Winder	Reinforcement Learning	O'reilly	ISBN-978-93–8588-950-9

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL736	Reinforcement Learning- Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E737	Edge AI			
Teaching Scheme (Hrs./Week)	**TH**	**P**	**TUT**	**Total**
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	**Marks**				
	LAB/TUT CA	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
	--	20	30	50	100

Course pre-requisites: Programming Fundamentals, Machine Learning Basics, Digital Electronics & Computer Architecture, Operating Systems (Basic Concepts), Embedded Systems

Course Objectives:

1. To understand the fundamentals of Edge AI and embedded machine learning systems.
2. To design and deploy lightweight ML models on resource-constrained embedded devices.
3. To implement TinyML workflows using microcontrollers and edge hardware platforms.
4. To develop real-world Edge AI applications integrating sensors, ML models, and embedded systems.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Analyze the architecture and constraints of Edge AI systems.
CO2. Develop and optimize machine learning models for embedded deployment.
CO3. Deploy TinyML models on microcontrollers using TensorFlow Lite for Microcontrollers.
CO4. Design and implement an end-to-end Edge AI solution for a real-world application.

Detailed Curriculum

Module No.	Unit No.	Contents	No of Hrs.	CO
1	Introduction to Edge AI and TinyML		09	CO1
	1.1	Evolution of AI: Cloud AI vs Edge AI Introduction to Edge Computing Need for Edge AI TinyML ecosystem		
	1.2	Embedded systems fundamentals Overview of microcontrollers (ARM Cortex-M) Sensors and data acquisition basics Power and memory constraints in edge devices		
2	Machine Learning for Embedded Systems		09	CO1 CO2
	2.1	ML workflow for embedded systems Data collection and preprocessing Feature extraction techniques		
	2.2	Model selection for constrained environments Quantization and model compression		
	2.3	Introduction to TensorFlow Lite TensorFlow Lite for Microcontrollers (TFLM)		
	2.4	Algorithms for Edge AI.		
		Self-reading: How to Build a Dataset		
3	Deploying AI Models on Edge Devices		09	CO2 CO3
	3.1	Model deployment pipeline Firmware development for ML inference Memory management strategies		
	3.2	Real-time inference on microcontrollers Optimization for latency and power Debugging embedded ML systems		
	3.3	The Hardware of Edge AI		
4	Edge AI Applications and Optimization		09	CO2 CO3 CO4
	4.1	Keyword spotting systems Gesture recognition		
	4.2	Predictive maintenance Anomaly detection		
	4.3	Power optimization techniques Model pruning and compression Security considerations in Edge AI		
	4.4	Tools and Expertise. Understanding and Framing Problems		
	4.5	Evaluating, Deploying, and Supporting Edge AI Applications		
5	Advanced Edge AI		09	CO3 CO4
	5.1	Edge AI in IoT ecosystems Edge AI in Healthcare, Smart Cities, Agriculture Integration with cloud services (hybrid architecture) Edge AI lifecycle management Ethics and privacy in Edge AI		
	5.2	Use Case: Wildlife Monitoring		
	5.3	Use Case: Food Quality Assurance, Use Case: Consumer Products.		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Daniel Situnayake & Jenny Plunkett	AI at the Edge	O'Reilly	January 2023: First Edition
2	Lucas Jennings	The AI Edge: Unlocking Profits with Artificial Intelligence	Atlantic	2024, first Edition
3	Pete Warden, Daniel Situnayake	TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers	Shroff/O'Reilly	2020 First Edition

*In addition to printed books, faculty can suggest (authentic) URLs or e-books, e-contents etc.

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL737	Edge AI Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Please note:

1. The total marks assigned for Continuous Assessment (LAB/TUT) as per scheme can be distributed in a number of components as given above.
2. Course coordinator should decide the rubrics/distribution of marks for different components in consultation with all other faculty teaching the same course.
3. The rubrics/distribution should be uniform for all batches.
4. The rubrics/distribution should be communicated to all students at the beginning of the semester.

Departmental Elective IV

Course Code	Name of the Course
216U01E741	Blockchain Technology

Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Networking Concepts, Object Oriented Programming Skills, Cryptography and Network Security Concepts.

Course Objectives:

The objective of the Course is to explore the Blockchain technology by exploring Bitcoin protocol followed by the Ethereum protocol – to lay the foundation necessary for developing applications and programming. Course will give the idea about the decentralized peer-to-peer network, an immutable distributed ledger and the trust model that defines a blockchain.

This course explains basic components of a blockchain (transaction, block, block header, and the chain) its operations (verification, validation, and consensus model) underlying algorithms, and essentials of trust (hard fork and soft fork).

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1 Describe the basic concepts of Blockchain and Distributed Ledger Technology
CO2 Apply cryptographic hash required for blockchain.
CO3 Categorize and discuss the consensus in Blockchain.
CO4 Infer the components of Ethereum ecosystem.
CO5 Design a private blockchain platforms

Module No.	Unit No.	Contents	Hrs.	CO
1	**Introduction to Blockchain**		**08**	**CO1**
	1.1	Challenges Faced by Modern Businesses, what is Blockchain?, Building Blocks of Blockchain, Types of Blockchain,		
	1.2	Introduction to Blockchain Pillars, Cryptography, Consensus, Distributed Ledger:		
	1.3	A Block in a Blockchain: Structure of a Block, Block Header , Block Identifiers: Block Header Hash and Block Height , The Genesis Block, Linking Blocks in the Blockchain.		
2	**Bitcoin Blockchain**		**12**	
	2.1	Introduction to Bitcoin, Bitcoin Wallets, Bitcoin Block, Bitcoin Transaction, Bitcoin Scripts, Bitcoin Attacks, Bitcoin Network, Bitcoin Mining.		**CO2**
	2.2	**Basic Crypto Primitives:** Cryptographic Hash Function, Properties of a hash function, Hash pointer and Markel tree, Digital Signature, Public Key Cryptography, A basic cryptocurrency.		
3	**Consensus and Mining**		**10**	**CO3**
	3.1	Problem, Independent Verification of Transactions, Mining Nodes, Aggregating Transactions into Blocks, Constructing the Block header, Mining the Block, Successfully Mining the Block, validating a New Block, Assembling and Selecting Chains of Blocks, to different consensus algorithms Proof of Work and Proof of Stake, PBFT.		
	3.2	**Working with Consensus in Bitcoin**: Distributed consensus in open environments, Consensus in a Bitcoin network, Proof of Work (PoW) – basic introduction, Hashcash PoW, Bitcoin PoW, Attacks on PoW and the monopoly problem, Proof of Stake, Proof of Burn and Proof of Elapsed Time, The life of a Bitcoin Miner, Mining Difficulty, Mining Pool.		
4	**Public / Ethereum Blockchain**		**08**	**CO4**
	4.1	Ethereum components: miner and mining node, Ethereum virtual machine, Ether, Gas, Transactions, accounts, Solidity and smart contract development.		
	4.2	Introduction to Swarm and whisper, Remix IDE, Truffle Framework, Ethereum Networks, Ethereum Wallets Ethereum Clients, Web3.js, NFT.		
5	**Private Blockchain**		**08**	**CO5**
	5.1	Introduction, Key characteristics, Need of Private Blockchain, Smart Contract in a Private Environment, State Machine Replication, Consensus Algorithms for Private		

		Blockchain - PAXOS and RAFT, Byzantine Faults: Byzantine Fault Tolerant (BFT) and Practical BFT.		
	Applications of Blockchain		04	
6	**6.1**	Blockchain in IOT, banking and Finance, Government, Healthcare system, AI, Blockchain in Education, Energy, Supply chain, Real-state.		CO1, 2,5
		Total	**45**	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Andreas M. Antonopoulos	Mastering the Bitcoin: Programming the Open Blockchain	O' Reilly	2nd Edition, 2017
2.	Melanie Swan	BlockChian	O'Reilly	2015
3.	Nitin Gaur, Luc Desrosiers, Petr Novotny, Venkatraman Ramakrishna	Hands-On Blockchain with Hyperledger: Building decentralized applications with Hyperledger Fabric and Composer	Packt	Kindle Edition, 2018
4.	*Stephen Fleming business ecosystems*	*Blockchain Technology: Introduction to Blockchain Technology and its impact on*	Stephen Fleming	2017
5.	*Zeeshan-ul- hassan Usmani*	*Introduction to lockchain with Case Studies*	Guhftgu Publication	2018

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL741	Blockchain Technology Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E742	Computer Vision			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Programming skills, as well as familiarity with linear algebra and knowledge of basic image processing fundamentals.

Course Objectives:

1. Introduce computer vision challenges and areas of applications.
2. Provide knowledge of various segmentation, feature detection and tracking algorithms.
3. Develop skills to extract 3-D information from single, two, and multiple views.
4. Provide knowledge motion detection & tracking.
5. Develop skills for Matlab / Open CV programming languages to design and develop Computer Vision applications.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Describe model perspective image formation used for calibrating single and multiple camera systems.

CO2. Design 3D position and shape information from arbitrary viewpoints.

CO3. Apply the problems in finding corresponding features in different viewpoints.

CO4. Analyse visual motion to recover scene structure and viewer motion, and understand how this information can be used for navigation.

CO5. Develop real-world applications and appreciate the industrial potential of computer vision by using various computer vision techniques.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Computer Vision**		03	CO1
	1.1	Introduction to computer vision, CV Challenges, Recognition Methodology and applications.		
2	**Image Formation: Geometry and Photometry**		12	CO2
	2.1	Photometric image formation: Lighting, Reflectance and shading, Optics.		
	2.2	Binary Machine Vision: Connected Component Labelling, Region Properties, Area and Shape parameters		
3	**Segmentation and Feature Extraction**		12	CO3
	3.1	Feature detectors, Feature descriptors for points and patches. Co-occurrence matrix and Texture features' detection, Introduction to feature detectors SIFT, SURF and HoG		
	3.2	Active contours, Split and merge, Mean shift and mode finding		
4	**Multi view Geometry & Motion Analysis**		10	CO4
	4.1	Geometric primitives and transformations: Geometric primitive's 2D transformations, 3D transformations, 3D rotations, 3D to 2D projections.		
	4.2	Shape from stereo: feature matching (Cross correlation technique), 3D Reconstruction: Shape from X Study topic: Structures from Motion: Motion detection and tracking, optical flow		
5	**Object Recognition and Understanding**		08	CO5
	5.1	Object detection, Face recognition, Instance recognition, Category recognition		
	5.2	Context and scene understanding.		
		Total	45	--

Reference Books

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Robert Haralick and Linda Shapiro	*Computer and Robot Vision", Vol. I, II,*	Addison-Wesley	Edition 1, 1993
2.	Richard Szeliski	*Computer Vision: Algorithms and Applications*	Springer	Edition 2, 2010
3.	Forsyth and J. Ponce	*Computer Vision: A Modern Approach, D.*	Prentice-Hall	Edition 2, 2011.
4.	R. Jain, R. Kasturi, B. G.Schunck	*Machine Vision*	McGraw Hill	Edition 1, 1995

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL742	Computer Vision-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E743	Advanced Web Technology			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

*** The ISE and End Semester Exam is conducted in Lab for with Onscreen Mode with guidelines and instructions provided during/before the examination.**

Course prerequisites (if any):
Fundamentals of Object Oriented Programming concepts.

Course Objectives:

1. Creating Form based application using WPF and .net Controls.
2. Creating ASP.Net applications using standard .net controls.
3. Connecting to data sources and managing them.
4. Understand the fundamentals of developing modular application by using object oriented methodologies

Course Outcomes:

At the end of successful completion of the course the student will be able to

CO1	Understand .NET framework & fundamentals.
CO2	Utilize the .NET framework to build distributed enterprise applications.
CO3	Implement ADO.NET and LINQ concept along with ASP.NET
CO4	Develop web applications using a combination of client-side (JavaScript, HTML, XML, WML) and server-side technologies (ASP.NET, ADO.NET).
CO5	Develop ASP.NET Web Services, secure web services, and .NET remoting applications

Module No.	Unit No.	Details	Hrs.	CO
1	**Introduction to .NET technologies**		**05**	**CO**
	1.1	Features of .NET, .NET Framework, CLR, What is ASP.NET? Difference between ASP and ASP.NET. Design View, HTML View, Default Files used in ASP.NET		**1**
	1.2	Concept of Master pages, Intrinsic Objects of ASP.Net, Structure of ASP.NET page, Cascading Style Sheet: Embedded, Inline, External.		
2	**Controls in ASP.NET**		**12**	**CO**
	2.1	Overview of Dynamic Web page, Understanding ASP.NET Controls, Applications, Web servers, Installation of IIS. Web forms, web form controls -server controls, client controls		**2**
	2.2	Adding controls to a web form, Buttons, Text Box, Labels, Checkbox, Radio Buttons, List Box. Adding controls at runtime. Running a web Application, creating a multiform web project. Form Validation.		
	2.3	Client side validation, server Side validation, validation Controls: Required Field Comparison Range.		
		Self-Learning: Calendar control, Ad rotator Control, Internet Explorer Control.		
3	**Overview of ADO.NET and XML**		**13**	**CO**
	3.1	ADO.NET Fundamentals: Understanding Databases, Configuring Your Database, Understanding SQL Basics, Understanding the Data Provider Model, Using Direct Data Access, Using Disconnected Data Access.		3
	3.2	Data Binding: Introducing Data Binding, Using Single-Value Data Binding, Using Repeated-Value Data Binding, Working with Data Source Controls, The Data Controls: The GridView, Formatting the GridView, Selecting a GridView Row, Editing with the GridView, Sorting and Paging the GridView, Using GridView Templates, The DetailsView and FormView.		
	3.3	LINQ Architecture, LINQ to object, LINQ to SQL, LINQ to Dataset		
4	**ASP.NET Applications**		**05**	**CO**
	4.1	Creating, tracking, caching, error handling, Securing ASP.NET applications.		4
	4.2	Form based applications, window based application. .NET Remoting.		
5	**Web services**		**10**	CO

	5.1	Introduction, State management- View state, Session state, Application state, Building ASP.NET web services, working with ASP.NET applications, creating custom controls, Invoking COM/COM+, ActivX Components.		5
	5.2	Deployment of ASP.NET application		
	5.3	Web services security Authentication & Authorization (Basic Auth, OAuth, JWT) Transport-level security (HTTPS/TLS) Message-level security (WS-Security)		
	5.4	Self-learning : Case studies of real-world Web Services (e.g., Amazon, Google APIs, Payment gateways)		
		Total	**45**	

Reference Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Albert S. Tanure	ASP.NET Core 9 Essentials	Packt Publishing, UK	1st Edition, 2025
2.	Mark J. Price	C# 13 and .NET 9 – Modern Cross-Platform Development Fundamentals	Packt Publishing Ltd, UK	9th Edition, 2024
3.	Jon Galloway, et al.	Professional ASP.NET Core 6	Wrox Press, USA	1st Edition, 2022
4.	Adam Freeman	Pro ASP.NET Core MVC 2	Apress, USA	1st Edition, 2017

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL743	Advanced Web Technology- Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E744	DevOps			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

1. Basic knowledge of Linux/Unix command-line operations.
2. Familiarity with at least one programming language (e.g., Python, Java).
3. Understanding of software development life cycle (SDLC).
4. Basic understanding of networking and system administration is helpful but not mandatory.

Course Objectives:

1. Understand the core principles, practices, and cultural aspects of DevOps.
2. Gain hands-on experience with tools and processes involved in version control, CI/CD, containerization, and infrastructure as code.
3. Develop and deploy applications using modern DevOps practices and pipelines.
4. Learn to automate and monitor application infrastructure using tools like Terraform, Jenkins, Docker, and Kubernetes.
5. Integrate testing in continuous integration and delivery pipelines.
6. Explore the differences and synergies between DevOps, ITIL, and Site Reliability Engineering (SRE).

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1: Understand core DevOps principles, CI/CD concepts, and version control systems.
CO2: Develop and manage CI/CD pipelines using tools like Git and Jenkins.
CO3: Apply containerization using Docker for application build and deployment.
CO4: Deploy and orchestrate applications using Kubernetes and implement IaC tools.
CO5: Monitor, troubleshoot, and optimize systems for performance, scalability, and security.

Module No.	Unit No.	Contents	No of Hrs.	CO
1		**DevOps Fundamentals and Linux Environment**	09	1
	1.1	Introduction to DevOps and SDLC DevOps lifecycle and practices Introduction to Operating Systems Virtualization & Virtual Machines (VMWare, VirtualBox) Linux File System and CLI		
	1.2	Basic Linux Commands, Pipes & Redirects Linux Package Managers, Vim Editor Users, Groups, Permissions & Ownership Environment Variables and SSH		
2		**Version Control and Build Automation**	09	2
	2.1	Introduction to Version Control and Git Git setup (local & remote repositories) Git operations: commit, branch, merge, rebase, stash Merge conflicts and history management Git for DevOps workflows		
	2.2	Introduction to Build Tools Build and run applications (Java/JS) Artifact creation and publishing Integration of Build Tools with Docker		
3		**Cloud, Artifact Management and Containerization**	09	3
	3.1	Introduction to Cloud Computing (IaaS) Server setup and deployment on cloud (e.g., Linode) Linux user configuration on cloud Artifact Repository (Nexus): setup, repository types, REST API Blob store, cleanup policies		
	3.2	Introduction to Containers and Docker Docker architecture, commands, volumes, networking Dockerfile and Docker Compose Private Docker repositories and deployment		
4		**CI/CD, Orchestration and Infrastructure Automation**	09	4
	4.1	Jenkins: installation, UI, jobs, pipelines, Jenkinsfile CI/CD pipeline creation and automation Webhooks and multibranch pipelines Kubernetes architecture and components Pods, Deployments, Services, Ingress ConfigMaps, Secrets, Volumes, StatefulSets		
	4.2	Helm and microservices deployment AWS EKS and container services Terraform: provisioning infrastructure (EKS, resources) Ansible: configuration management, playbooks, deployment Integration: Jenkins + Terraform + Ansible		
5		**Monitoring, Logging and DevOps Practices**	09	3 4

	5.1	Monitoring with Prometheus Grafana dashboards and visualization Alert rules and Alertmanager configuration Email alerting and notifications Monitoring third-party applicationsEthics and privacy in Edge AI		
	5.2	Troubleshooting pipelines and deployments DevSecOps basics and security practices Performance optimization and scalability Best practices in DevOps lifecycle		
		Total	45	--

Reference Books

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Len Bass, Ingo Weber, Liming Zhu	*DevOps: A Software Architect's Perspective*	Addison-Wesley	2nd Edition / 2021
2	Gene Kim, Jez Humble, Patrick Debois, John Willis	*The DevOps Handbook: How to Create World-Class Agility, Reliability, & Security*	IT Revolution Press	2nd Edition / 2021
3	Jez Humble, David Farley	*Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation*	Addison-Wesley	1st Edition / 2010
4	Docker Inc. (Community Authors)	*Docker Deep Dive*	Independently Published	Latest (e-book)
5	Kelsey Hightower, Brendan Burns, Joe Beda	*Kubernetes: Up and Running*	O'Reilly Media	3rd Edition / 2022
6	https://docs.github.com/ – GitHub Documentation https://www.jenkins.io/doc/ – Jenkins Official Docs https://kubernetes.io/docs/ – Kubernetes Documentation https://learn.hashicorp.com/terraform – Terraform Tutorials https://docs.gitlab.com/ee/ci/ – GitLab CI/CD Docs https://www.selenium.dev/documentation/			
7	DevOps by Prof. Rajiv Misra, IIT Patna NPTEL DevOps Course Software Engineering (includes DevOps practices) by Prof. Rajib Mall, IIT Kharagpur NPTEL Software Engineering.			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL744	DevOps- Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E745	Natural Language Processing			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basic Probability theory, basics of machine learning fundamentals and Python programming language.

Course Objectives:

1. To provide a strong foundation in Natural Language Processing (NLP).
2. To develop an understanding of syntax analysis and parsing techniques.
3. To explore word representations and semantic modeling.
4. To understand neural language models and deep learning techniques in NLP
5. To dive into modern transformer-based models and pretrained language models.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1: Understand the core concepts, techniques in text analysis, and challenges in natural language processing.

CO2: Apply language modeling and part-of-speech tagging, and analyze syntactic parsing techniques to extract sentence structures for text analysis and understanding.

CO3: Utilize word representation models to capture semantic meaning and improve language understanding.

CO4: Understand and apply neural language models, including RNNs, LSTMs, and attention mechanisms, for various sequence-based NLP tasks.

CO5: Design and apply various NLP models and techniques to build real-world natural language applications.

Detailed Curriculum

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction & Word level analysis**		08	CO1
	1.1	NLP introduction, levels, ambiguity in natural language, stages in NLP and applications		
	1.2	Basic Text Processing Techniques: Tokenization, Sentence Segmentation, Lemmatization, Edit distance for spell correction & computational Morphology Concepts.		
	1.3	Part of Speech Tagging Problem in NLP, Hidden Markov Model for POS Tagging.		
	1.4	Self-learning: Porter stemmer algorithm for stemming.		
2	**Language Modeling & Syntax Analysis**		08	CO2
	2.1	N-Gram language models, evaluation of language models (Perplexity) and smoothing techniques.		
	2.2	Syntax – Introduction, Constituency, Constituent Phrases & Context-free Grammar. Parsing, Dynamic programming, CKY algorithm & PCFGs.		
	2.3	Introduction to dependency grammar, transition-based parsing methods, and their application in NLP.		
	2.4	#Self-Learning: Attachment for fragment of English- sentences, noun phrases, Verb phrases and prepositional phrases.		
3	**Distributional Semantics and Word Representation**		10	CO3
	3.1	The distributional semantic (Vector space) model. Lexical semantics Relations among lexemes & their senses, WordNet, word sense disambiguation problem in nlp.		
	3.2	Word vectors, Word2vec - a distributed representation, word embedding, learning word vectors CBOW & skip gram models.		
	3.3	Glove & FastText embeddings.		
	3.4	#Self-Learning: Multilingual representations with emphasis on Indian Languages		
4	**Neural Language Models**		09	CO4
	4.1	RNN language model and LSTM.		
	4.2	LSTMs for Sequence Labeling, LSTMs for Sequence to Sequence task.		
	4.3	Attention in Sequence-to-Sequence Models		
	4.4	Self-reading: GRU and Bi-LSTMs		
5	**Introduction to Transformer Models**		10	CO5
	5.1	Motivation for transformers (limitations of RNNs/LSTMs), Self-attention mechanism, Multi-head attention, Positional encoding & layer normalization.		
	5.2	Transformer architecture (Encoder-Decoder overview), Decoding strategies (Greedy decoding and Beam search).		
		Language translation and summarization applications using basic encoder-decoder models.		

	5.4	Self-reading: Pretrained Bi-LSTMs: ELMO		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Daniel Jurafsky and James H. Martin.	Speech and Language Processing	Prentice Hall	Third Edition/2024
2	Jacob Eisenstein	Natural Language Processing	The MIT Press,	First edition/2019
3	Christopher D. Manning and Hinrich Schutze	Foundations of Statistical Natural Language Processing	MIT Press	1999
4	Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana	Practical Natural Language Processing	O'Reilly Media, Inc	2020
5	Hobson Lane, Cole Howard, Hannes Hapke	Natural Language Processing in Action	Manning Online	2019
6	Weblinks: 1. liveBook · Manning 2. Speech and Language Processing			
7	NPTEL course: Natural Language Processing - Course Deep Learning for Natural Language Processing - Course			
8	Coursera Course: Natural Language Processing | Coursera			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL745	Natural Language Processing-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E746	Secure Coding			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:
Knowledge of programming languages, cryptography, application development.

Course Objectives:

1. Understand the fundamental principles of secure software development and common sources of software vulnerabilities.
2. Identify, analyze, and mitigate security flaws in code through the application of secure coding practices.
3. Utilize security frameworks, standards, and tools (e.g., OWASP, CERT) for code analysis and vulnerability assessment.
4. Integrate security into development processes for resilient application development.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Use the fundamental principles of secure software development and identify common programming vulnerabilities

CO2. Analyze and evaluate insecure code segments to detect potential security flaws.

CO3. Design and implement software systems that integrate security measures throughout the development lifecycle.

CO4. Apply secure coding standards, frameworks, and tools (e.g., OWASP, CERT) to develop secure software solutions

CO5. Assess software for compliance with security best practices and recommend improvements using Security frameworks and standards.

Detailed Contents

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Secure Software Development.**		07	CO1
	1.1	Definition and importance of software security, Difference between software quality and software security, Common security goals: Confidentiality, Integrity, Availability (CIA triad), Secure vs. insecure software characteristics.		
	1.2	Security in the Software Development Lifecycle (SDLC), Phases of SDLC and security integration points, The concept of "Shift Left" in security, Secure SDLC models: Microsoft SDL, OWASP SAMM, BSIMM, Role of developers and security champions in SDLC		
		Self-learning: Overview of threat modeling and risk assessment, Introduction to security guidelines (OWASP, CERT, CWE, ISO/IEC 27034)		
2	**Software Vulnerabilities & Challenges**		12	CO2
	2.1	Input Validation and Data Sanitization, Importance of validating user input, Whitelisting vs. blacklisting approaches, Common input handling mistakes, Data sanitization techniques for different contexts (HTML, SQL, command line), Case studies on vulnerabilities caused due to improper validation.		
	2.2	Memory and Buffer Vulnerabilities, Understanding memory layout and stack/heap concepts, Buffer overflow and underflow vulnerabilities, Format string vulnerabilities, Stack smashing and heap spraying attacks, Techniques for prevention: bounds checking, safe libraries, and compiler protections.		
	2.3	Injection-Based Attacks: SQL injection, command injection, and code injection mechanisms, Impact and exploitation of injection flaws, Secure coding techniques to prevent injections (parameterized queries, ORM use), Role of input validation and encoding. Web Application Vulnerabilities: Cross-Site Scripting (XSS): types, impact, and prevention, Cross-Site Request Forgery (CSRF): mechanism and defense, Directory traversal and file inclusion vulnerabilities, Session management and cookie security issues		
	2.4	Authentication, Authorization, and Session Management Flaws, Weak password storage and handling, Broken authentication and session fixation, Privilege escalation and improper access control, Secure password management and token-based authentication, Implementing multi-factor authentication (MFA)		
		#Self-Learning: Cryptographic Vulnerabilities		
3	**Secure Coding Practices**		12	CO3
	3.1	Principles of Secure Coding, Overview of secure coding philosophy, Difference between functional correctness and security correctness, Security design principles (least privilege, fail-safe defaults, economy of mechanism, complete mediation), Coding standards and best practices (CERT, OWASP, MISRA), Importance of code readability, maintainability, and documentation in secure development		

	3.2	Defensive Programming and its need, Input validation, exception handling, and fault tolerance, Assertions and sanity checks, Logging and auditing best practices, Graceful degradation and error recovery techniques, Memory Management and Pointer Safety, Safe memory allocation and deallocation in C/C++, Avoiding dangling pointers, memory leaks, and double frees, Secure use of dynamic memory and arrays, Tools for detecting memory errors (Valgrind, Address Sanitizer), Secure string handling and buffer operations		
	3.3	Secure Use of APIs and Libraries, Risks associated with third-party and open-source libraries, API misuse and dependency management vulnerabilities, Safe handling of file I/O, network APIs, and system calls, Sandboxing and limiting external dependencies, Secure configuration and dependency scanning		
		#Self-Learning: Case Studies in Secure and Insecure Programming (Heartbleed, Shellshock).		
4	**Security Tools and Code Analysis**		07	CO4
	4.1	Introduction to Code Analysis, Overview of code analysis in secure development, Difference between static and dynamic analysis, Role of automated tools in identifying vulnerabilities, Benefits and limitations of automated analysis, Integrating analysis tools in SDLC		
	4.2	Static Code Analysis Tools and Techniques, Principles of static code analysis, Common tools: SonarQube, Flawfinder, Fortify, Checkmarx, CodeQL, Detecting common issues: buffer overflows, uninitialized variables, injection points, Best practices for using static analysis in CI/CD environments		
	4.3	Dynamic Analysis and Fuzz Testing, Concept of runtime (dynamic) analysis, Fuzz testing fundamentals and workflow, Popular fuzzing tools (AFL, Peach Fuzzer, LibFuzzer), Debugging and runtime monitoring techniques, Memory leak and race condition detection Secure Code Review Process, Steps in a secure code review, Automated vs. manual code review approaches.		
		Self-reading: Vulnerability Scanning and Penetration Testing Tools (Nikto, OpenVAS, Nessus, OWASP, Burp Suite).		
5	**Principles of Secure Software Design**		07	CO5
	5.1	Core principles: least privilege, defense in depth, fail-safe defaults, open design, separation of duties, Secure architecture vs. secure implementation, Common design flaws and their mitigation strategies, Case study: Secure design in web and enterprise applications		
	5.2	Security in Software Development Lifecycle (SDLC), Embedding security across SDLC phases: requirements, design, coding, testing, deployment, maintenance, Secure SDLC models: Microsoft SDL, OWASP SAMM, BSIMM, Threat modeling and risk management during design, Role of security reviews and audits in SDLC, Shift-left security approach and automation		
	5.3	Compliance, Standards, and Legal Frameworks, Overview of software and data security compliance: ISO 27001, NIST, GDPR, HIPAA, Ethical and legal aspects of secure coding		

		Self-reading: Security in Agile and DevOps Environments (DevSecOps) 'The $440 Million Software Error at Knight Capital'		
			Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Robert C. Seacord	Secure Coding in C and C++	Addison-Wesley	2nd Edition/2013
2	Mark G. Graff and Kenneth R. van Wyk	Secure Coding: Principles and Practices	O'Reilly	2003
3	Michael Howard and David LeBlanc	Writing Secure Code	Microsoft Press	2nd Edition
4	Mark Merkow, Lakshmikanth Raghvan	Secure and Resilient software: Requirements, test cases and testing methods	CRC Press	2012
5	Neil Daswani, Christoph Kern, and Anita Kesavan.	Foundations of Security	Apress	2007
6	**OWASP Secure Coding Practices Guide** https://owasp.org/www-project-secure-coding-practices/			
7	**CERT Secure Coding Standards (Carnegie Mellon University SEI)** https://wiki.sei.cmu.edu/confluence/display/seccode			
8	**MITRE Common Weakness Enumeration (CWE)** https://cwe.mitre.org/			
9	**ISO/IEC 27034 – Application Security Overview** https://www.iso.org/standard/44378.html			
10	**NIST Software Assurance Resources** https://samate.nist.gov/			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL746	Secure Coding-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E747	Software Architecture			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Students should be familiar with basic concepts of Software Engineering

Course Objectives:

1. Develop architectural thinking applicable in building large and complex systems.
2. Structural and behavioural models in design notations such as Architecture description languages (ADLs).
3. Analysing the role of architectures in various environments.

Course Outcomes (CO):

At the end of the successful completion of the course, the student will be able to

CO1. Design the architecture of software systems in various architectural styles.

CO2. Evaluate software architecture using analysis techniques.

CO3. Analyze non-functional properties in the architectural design.

CO4. Design domain models for domain-specific Software Engineering and Architecture

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction of Software Architecture**		10	CO1
	1.1	Basic Concepts, Designing Architectures, Conventional Architectural styles : Concepts of Software Architecture, Models, Processes, Stakeholders		
	1.2	Styles and Architectural Patterns, Pipes and Filters, Event- based		
	1.3	Implicit Invocation, Layered systems, Repositories, Interpreters		
	1.4	**#Self-learning:** Architecture Design Process, Architectural Conception		
2	**Connectors :**		09	CO1
	2.1	Connector Foundations, Connector Roles		
	2.2	Connector Types and Their Variation Dimensions		
3	**Modeling , Analysis :**		10	CO2
	3.1	Modelling Concepts,Ambiguity, Accuracy,and precision.		
	3.2	Complex Modeling: Mixed Content and Multiple Views. Analysis Goals, Scope of Analysis, Architectural Concern being		
	3.3	Analyzed, Level of Formality of Architectural Models, Type of Analysis, Analysis Techniques.		
	3.4	#Self-learning::SAAM.		
4	**Designing for Non-Functional Properties and Implementation.**		08	CO3
	4.1	Efficiency, Complexity, Scalability, Heterogeneity, Adaptability, Dependability		
	4.2	#Self-learning: Study of Existing Frameworks		
5	**Domain-Specific Software Engineering.**		08	CO3 CO4
	5.1	Domain-Specific Software Engineering in a Nutshell, Domain-Specific Software Architecture, DSSAs.		
	5.2	#Self-learning: Product Lines, and Architectural Styles		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Richard N. Taylor, Nenad Medvidovic, Eric Dashofy	, "Software Architecture: Foundations, Theory, and Practice"	**Publisher**: Wiley. ISBN: 978-0-470-16774-8	10th Edition 2010
2	M. Shaw	"Software Architecture Perspectives on an Emerging Discipline"	Prentice-Hall of India Pvt.Ltd	1st Edition 1996
3	Len Bass, Paul Clements, Rick Kazman	Software Architecture in Practice	O Reilly	3rd Edition

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL747	Software Architecture- Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Course Title
216U01P701	Project II

	TH	P	TUT	Total
Teaching Scheme(Hrs.)	--	10	--	10
Credits Assigned	--	05	--	05

Examination Scheme	Marks							
	CA		ESE	TW	O	P	LabCA	Total
	ISE	IA						
	--	--	--	--	--	--	100	100

For Students opting for one semester long internship in semester VII, Project I will not continue to Project II.

Course Outcomes:

After successful completion of the course the student will be able to :

CO1. Implement and test the hardware / software components of the system to meet desired specifications
CO2. Analyze, interpret results and correspondingly modify the designed system/ components to achieve desired results.
CO3. Identify limitations of work done and suggest scope for future development.
CO4. Write technical report and present the work done in appropriate format.
CO5. Demonstrate individual, group and leadership skills during project execution

Term work:

The final year students have already under gone project assignment in their sixth semester and in this semester the students are expected to continue the project work of stage I. In addition to this they are required to report to the faculty advisor about the progress of the work on regular basis. There will be an intermediate evaluation at the mid of the semester by a panel of examiners. A continuous assessment record of the progress of the project will be maintained by concerned faculty advisor.

The final evaluation will be done by a pair of external examiner and internal examiner. The evaluation will include; a presentation by the group members on the project along with demonstration of the work done and oral examination. Each individual student will be assessed for his/her contribution, understanding and knowledge gained during the examination.

Semester VIII

Departmental Elective -V

Course Code	Name of the Course			
216U01E751	Advanced Algorithms: Design & Analysis (AADA)			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	03
Credits Assigned	02	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Programming Skills

Course Objectives:

1. To study algorithms for solving practical problems efficiently along with their theoretical performance analysis.
2. Understand the foundations of complexity theory as a formal study of algorithm performance.
3. To explore major categories of algorithms including greedy, dynamic programming, network flow, graph, intelligent, and approximation algorithms.
4. Build a foundation for advanced study in computer algorithms and their performance.
5. Develop both theoretical knowledge and practical problem-solving skills using diverse algorithmic strategies.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1	Understand and derive the time and space complexity of algorithms using various analytical techniques.
CO2	Solve path computation and network flow problems using various approaches
CO3	Apply appropriate data structure and algorithm design techniques to develop efficient solutions for various problems.
CO4	Analyze performance of randomization and intelligent algorithms over traditional solutions

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to analysis of Algorithms**		06	CO1
	1.1	Design and Analysis Fundamentals, Performance analysis, space and time complexity, classes of Problems		

	1.2	Amortized Analysis: Aggregate Method, Accounting Method, Potential Method		
	1.3	Algorithm analysis: Master's theorem, Substitution and Recursion tree methods		
2	**Advanced Data Structure**		08	CO2
	2.1	Introduction to trees and heap Red-Black Trees: properties of red-black trees, Operations on Red-black trees		
	2.2	Probabilistic Data Structure: LogLog and HyperLogLog, Count Min sketch.		
	2.3	Spatial Data Structure: KD Tree, R Tree		
		Self- Learning: Fibonacci heap, and its advantages over binomial heap and binary heap.		
3	**Graph algorithms**		06	CO3
	3.1	Path computation algorithms: Floyd Warshall shortest path algorithm, Johnson's method		
	3.2	Flow Networks: The Ford Fulkerson method, maximum bipartite matching, Push relabel algorithms, The relabel to front algorithm,Edmonds-Karp Method		
		#Self-Learning: All pairs shortest path by using Matrix Multiplication.		
4	**Approximation Algorithms**		10	CO4
	4.1	**Algorithm Classes:** P, NP, NP Hardness and NP Completeness **Randomized Classes:** RP, BPP, ZPP (Adleman's theorem)		
	4.2	**Np Completeness Proofs:** Satisfiability(3 sat), Reducibility, TSP.		
	4.3	**Approximation Algorithms:** Vertex Cover Problem, Travelling Sales Person problem. **Network Approximation:** Randomized Rounding, Primal Dual algorithms.		
		#Self learning : Turing Machine Halting Problem (time and space bounds, nondeterminism), Diagonalization problem.		
		Total	30	--

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1.	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest	*Introduction to Algorithms*	PHI, India	Second Edition
2	Rajeev Motwani, Prabhakar Raghavan,	*Randomized Algorithm*	Cambridge University Press	First Edition/ South Asia Edition
3	Peter Norvig and StuartJ. Russell	*Artificial Intelligence: AModern Approach*	Pearson Education	5th Edition
4	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein,	*Introduction to Algorithms*	PHI	2nd Edition
5	Horowitz, Sahani and Rajsekaran	*Fundamentals of Computer Algorithms*	Galgotia	2nd Edition
6	Harsh Bhasin,	*Algorithms – Design and Analysis‖,*	Oxford	1st Edition, 2015

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL751	**Advanced Algorithms: Design & Analysis (AADA)**			
Teaching Scheme (Hrs./Week)	**TH**	**P**	**TUT**	**Total**
	--	**02**	**--**	**02**
Credits Assigned	**--**	**01**	**--**	**01**

Evaluation Scheme	**Marks**				
	LAB/TUT CA*	**CA (TH)**		**ESE**	**Total**
		IA	**ISE**		
	50	**--**	**--**	**--**	**50**

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Bioinformatics			
216U01E752	Bioinformatics			
Teaching Scheme (Hrs./Week)	**TH**	**P**	**TUT**	**Total**
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basic concepts of Programming

Course Objectives:

1. To introduce students to the fundamentals of molecular biology and recent advances in genomics technology
2. The development of bioinformatics as an interdisciplinary field has introduced many sophisticated tools and techniques, to organize the information associated with biological molecules and contribute to our understanding of biological processes.
3. On a larger scale, this has led to many practical applications, not only providing greater depth to biological research but also adding other dimensions to engineering applications.
4. To provide students with a practical and hands-on experience with common bioinformatics tools and databases.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand the scope of Bioinformatics

CO2. Understand popular bioinformatics database

CO3. Learn Fundamentals of Databases and Sequence alignment

CO4. Process, analyze, and manage biological information through Machine Learning Algorithms.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Bioinformatics**		06	CO1
	1.1	Bioinformatics- History, Computational biology, Biological computing, Applications		
	1.2	Human Genome Project: History, Significance & Applications. Introduction to important bioinformatics databases (NCBI, Uniprot, PDB .		
	1.3	Self-learning:other bioinformation datasets		
2	**Algorithms for biological sequence analysis**		06	CO2
	2.1	Introduction to python for bioinformatics, Visualization of Genomics, Sequencing technologies: Overview, Genome Assembly, Genome Annotation, Comparing Genomes		
	2.2	Pairwise sequence analysis,		
	2.4	#Self-Learning: Multiple sequence analysis		
3	**Introduction to protein structure prediction**		06	CO3
	3.1	Predictive methods using protein sequences – protein identification, physical properties, motifs and patterns		
	3.2	3D structure prediction techniques, structure classification, Secondary structure prediction,		
	3.4	#Self-Learning:Tertiary structure prediction		
4	**Machine Learning for Bioinformatics**		6	CO4
	4.1	Intelligent systems, Hidden Markov model(HMM), Bayesian network(BN), Symbolic machine learning (Decision and identification trees),		
	4.2	Application of Artificial intelligence (AI) for bioinformatics, Artificial neural network (ANN), Stochastic context free grammar (SCFG), Genetic and Lamarckian algorithms.		
	4.3	Self-reading: Case studies (e.g., cancer classification, protein structure prediction), Application of Artificial intelligence(AI) for bioinformatics,		
5	**Applications of Bioinformatics**		06	CO4
	5.1	Secondary Structure Prediction of Protein,Secondary structure: basic principles on which the prediction methods of first, second and third generation are based		
	5.2	Algorithm of Chou-Fasman, GOR methods, predicting secondary structures using these methods and analysis		
	5.3	Concepts in measuring the accuracy of prediction (Q3, segment overlap, Mathew's correlation coefficient etc.)		
	5.4	#Self learning : Comparative and Functional Genomics		
		Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1.	Pierre Baldi	Bioinformatics: The Machine Learning Approach (Adaptive Computation and Machine Learning series	Bradford	2008
2.	Dan E. Krane	Fundamental Concepts of Bioinformatics	Pearson	2009
3.	Andreas Baxevanis	Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins	Wiley-Interscience	2001

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL752	Bioinformatics -Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E753	Business Analytics			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	20	30	50	150

Course pre-requisites: Engineering Maths and Excel

Course Objectives:

1. Understand the fundamental concepts and tools of business analytics and their role in organizational decision-making.
2. Apply managerial statistical tools in descriptive and predictive analytics, including trend line analysis, time series forecasting, risk analysis, simulation, and data mining.
3. Develop the ability to interpret and effectively use analytic models and results to support better business decisions.
4. Learn to communicate insights and collaborate with analytics professionals in a business environment.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand and perform Business analytics, Database analytics from a variety of business scenarios

CO2. Applying statistical inference techniques to conduct and interpret result of hypothesis testing

CO3. Use of Trendlines and Regression analysis to fit models to data

CO4. Understand forecasting model based on time series

CO5. Apply expected values in making the decision

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to business analytics**		5	CO1
	1.1	**1.1 Analytics Foundations** Introduction, Evolution of Business Analytics, Types (Descriptive, Predictive, Prescriptive), Analytical Models, Applications in various domains		
	1.2	**Data Foundations and Tools** Data sources, Data types, Data quality issues, Structured vs Unstructured Data Introduction to Excel: formulas, pivot tables, filtering, functions Overview of tools: Power BI / Google Data Studio basics, Python Pandas		
2	Descriptive Analytics		6	CO2
	2.1	Measures of central tendency and dispersion		
	2.2	Visualization: bar charts, line plots, histograms, dashboards		
	2.3	Statistical Inference: Hypothesis testing (one-sample, two-sample), ANOVA, Chi-square test		
	2.4	#Self-Learning: Case Study on Customer Care Analytics using Excel / Power BI		
3	**Quantum Optimization Algorithms**		6	CO3
	3.1	Trendlines and regression analysis, Simple linear regression, multiple linear regression ,		
	3.2	Building regression model with categorical variables, regression model with Non-linear terms		
	3.3	Regression error metrics		
	3.4	#Self-Learning: Case study on Revenue management		
4			6	CO4
	4.1	Qualitative and Quantitative Forecasting, Time Series Forecasting Models:Stationary series, Linear trend, Seasonality		
	4.2	Moving average, exponential smoothing, Regression-based forecasting with causal variables		
	4.3	Forecast error metrics (MAE, RMSE)		
	4.4	**Self-reading:** Case Study: Sales Performance Forecasting		
5	**Decision Analysis**		7	CO5
	5.1	Decision Making under Uncertainty, Decision Trees and Strategies		
	5.2	Utility and Expected Value, Sensitivity Analysis and Scenario Modeling		
	5.3	Optimization using Excel Solver, Introduction to Prescriptive Analytics		
	5.4	**Self-reading:**Case Study: Marketing Budget Allocation / Lending Decisions		
		Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	S. Christian Albright & Wayne L. Winston	Business Analytics: Data Analysis and Decision Making	Cengage Learning	8th Edition (2024)
2	James R. Evans	Business Analytics	Pearson	3rd Edition (2024)
3	Sandhya Kuruganti , Hindol Basu	Business Analytics Applications to Consumer Marketing	Mc, Graw Hill Education	1st Edition 2015
4	U. Dinesh Kumar	Business Analytics: The Science of Data-Driven Decision Making	Wiley India	2nd Edition, 2023
5	Sanjiv Jaggia, Alison Kelly, Kevin Lertwachara	Business Analytics	Mc, Graw Hill Education	Latest Global Edition, 2024
6	Routledge “Mastering Business Analytics” Series, https://www.routledge.com/Mastering-Business-Analytics/book-series/BUSAN			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL753	Business Analytics Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E754	Cyber-Physical Systems			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Software Engineering.

Course Objectives:

1. Understand basics of information security, crimes and defence.
2. Learn principles and process of digital forensics.
3. Analyse and interpret disk forensic evidence.
4. Use tools to extract and investigate network evidence.
5. Know legal aspects of cyber-crimes under IT Act.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand the basic principles of CPS
CO2. Identify the CPS components and relevant dynamical aspects
CO3. Develop an exposition of the challenges in implementing a cyber-physical system
CO4. Understand Intelligent CPS models and controls
CO5. Understand CPS security and safety aspects

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Advanced Cyber-Physical Systems**		05	CO1
	1.1	CPS in Real-World Scenarios: Smart grids, healthcare, transportation, From Industry 4.0 to Industry 5.0: Human-centric systems, cobots, personalization. Cyber-Physical-Social Systems (CPSS). Introduction to Digital Twins in CPS, Design Principles: Modularity, scalability, interoperability, Validation and Simulation: MATLAB/Simulink, Modelica		
2	**CPS Hardware-Software Platforms**		07	CO2
	2.1	Embedded Platforms for CPS: Raspberry Pi, Jetson Nano, STM32, Sensor Networks: Types, precision, calibration (e.g., LIDAR, MEMS IMUs), Actuators: Smart electrical, electromechanical, and hybrid actuators,		
	2.2	Software-Defined Vehicles (SDV): OTA updates, middleware stacks, Introduction to Hardware-in-the-Loop (HIL) Simulation		
3	**CPS Communication and Networking**		06	CO3
	3.1	Overview of Industrial Protocols: OPC UA, MQTT, DDS, Time Sensitive Networking (TSN), Wireless Standards: 5G, LPWAN (LoRaWAN, NB-IoT), Networked CPS: Interoperability, latency, reliability, Real-Time Operating Systems for CPS (RTOS): FreeRTOS, Zephyr		
4	**Control, Scheduling & Decision Systems in CPS**		07	CO4
	4.1	Principles of Dynamical Systems (Retained Core). Model Predictive Control (MPC) and Linear Quadratic Regulators (LQR), Feedback and Feedforward Control in CPS. Scheduling: Fixed, Dynamic, and AI-driven scheduling at Edge, Edge AI & TinyML for Real-Time Inference		
5	**Cyber Law**		05	CO5
	5.1	Security Challenges in CPS: Threat models, attack surfaces, Blockchain for CPS Communication Integrity, Federated Learning in CPS: Collaborative model training across nodes, Secure Firmware Update Protocols, Case Studies: Automotive CPS, Medical CPS vulnerabilities		
		Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	E. A. Lee, Sanjit Seshia	Introduction to Embedded Systems – A Cyber–Physical Systems Approach	MIT Press	2nd Edition, 2017
2	Rajeev Alur	Principles of Cyber-Physical Systems	MIT Press	Kindle Edition , 2015
3	Andr´e Platzer	Logical Analysis of Hybrid Systems: Proving Theorems for Complex Dynamics	Springer	Kindle Edition, 2010
4	Houbing Song, Danda B. Rawat, Sabina Jeschke, Christian Brecher	Cyber-Physical Systems: Foundations, Principles and Applications	Academic Press	1st Edition, 2016
5	Houbing Song, Glenn A. Fink, Sabina Jeschke	Security and Privacy in Cyber-Physical Systems: Foundations, Principles, and Applications	Wiley-IEEE Press	1st Edition, 2017
6	Walid M. Taha. Abd-Elhamid M.Taha John Thunberg	Cyber Physical System - A Model based Approach	Springer	1st Edition, 2021

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL754	Cyber-Physical Systems-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E755	Deep Learning			
Teaching Scheme (Hrs./Week)	**TH**	**P**	**TUT**	**Total**
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Linear Algebra, Calculus (vectors, matrices, basic integrals), Probability (Bayes theorem, expectation, variance) & Basic machine learning (linear models, Regression, decision trees)

Course Objectives:

This course covers the basics of machine learning, neural networks and deep learning. Models for deep learning techniques and the various optimization and generalization mechanisms are included. Major topics in deep learning and dimensionality reduction techniques are covered.

The objective of this course is:

1. To present the mathematical, statistical and computational challenges of building neural networks
2. To study the concepts of deep learning
3. To enable the students to know deep learning techniques to support Real-time applications
4. To examine the case studies of deep learning techniques

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Explain the fundamentals of deep learning, neural network architectures, optimization techniques, and regularization strategies.

CO2. Apply convolutional neural network (CNN) architectures such as LeNet, AlexNet, VGGNet, ResNet for image and text-based classification tasks.

CO3. Analyze and implement unsupervised deep learning models such as Autoencoders and Generative Adversarial Networks (GANs) for feature learning and data generation.

CO4. Design and evaluate sequence modelling approaches using RNNs, LSTMs, and GRUs for real-world applications like text, speech, and time-series analysis.

Module No.	Unit No.	Contents	No of Hrs.	CO

1	**Introduction to Deep Learning**		06	CO1
	1.1	Difference between deep learning and machine learning, Difference between shallow networks and deep networks		
	1.2	Learning rates and data Normalization, activation functions, Optimizers, Regularization, Dropout, Momentum and Batch Norm.		
	1.3	The Backpropagation Algorithm-Stochastic and Minibatch Gradient Descent – Test Sets – Validation Sets – and Overfitting- Preventing Overfitting in Deep Neural Network		
2	**Convolutional Neural Networks (CNN)**		08	CO1 CO2
	2.1	Convolutional Neural Networks: Motivation,Convolution operations, Pooling, stacking, and striding.		
	2.2	CNN Architecture: LeNet,AlexNet,GoogLeNet,VGGNet,ResNet		
	2.3	Applications like image, and text classification.		
	2.4	#Self-Learning:MobileNet		
3	**Deep Unsupervised Learning**		08	CO3
	3.1	Autoencoders: Types of Autoencoders: Vanilla Autoencoder,Sparse Autoencoder,Denoising Autoencoder, Convolutional Autoencoders (CAE),Variational Autoencoders (VAEs)		
	3.2	Generative Adversarial Networks (GANs): Types of GANs: Vanilla GAN:,Deep Convolutional GAN,Conditional Gan (Cgan),Laplacian Pyramid GAN (LAPGAN):		
	3.4	#Self-Learning: Super-Resolution GAN (SRGAN),Variational Autoencoders (VAEs):		
4	**Sequence Modeling**		08	CO4
	4.1	Motivation,Recurrent Neural Networks (RNNs), bidirectional RNNs, ,RNN Architectures,Applications of RNN		
	4.2	Vanishing/Exploding gradient problem		
	4.3	Long Short Term Memory(LSTM),Applications of LSTMGreatest Recurrent Unit (GRU.)		
	4.4	Self-reading: Applications of GRU		
		Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Ian Goodfellow, YoshuaBengio,Aaron Courville	Deep Learning	An MIT Press book	2016
2	Deng & Yu	Deep Learning: Methods and Applications	Now Publishers	2013
3	Michael Nielsen	Neural Networks and Deep Learning	Determination Press	2015
4	NPTEL Course: Deep Learning - IIT Ropar By Prof. Mitesh M. Khapra, Prof. Sudarshan Iyengar IIT Madras, IIT Ropar			
5	NPTEL Course: Deep Learning By Prof. Prabir Kumar Biswas | IIT Kharagpur			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL755	Deep Learning- Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E756	Internet of Everything			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	04
Credits Assigned	02	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Microcontroller and Embedded systems

Course Objectives:

1. Understanding the characteristics and conceptual framework of the Internet of Things (IoT) and its evolution into the Internet of Everything (IoE).
2. Exploring IoE Architectures at different levels of IoE architectures and understanding how smart objects connect to various access technologies.
3. Analyzing technologies involved in IoE, such as Arduino, Raspberry PI, wireless sensor networks, and cloud platforms.
4. Applying IoE concepts to real-world applications like smart cities, home automation, and healthcare monitoring.

Course Outcomes (CO):

At the end of the successful completion of the course, the student will be able to

CO1 Interpret the Internet of Things concepts and challenges and its evolution into IoE

CO2 Evaluate the software and hardware platforms for IoE Technologies

CO3 Analyze IoE application data using IoE Analytics.

CO4 Design and develop small IoE applications.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Internet of Everything**		04	CO1
	1.1	Overview of IoT and IoE		
	1.2	Architecture of IoE systems		
	1.3	IoE Enabling Technologies		
2	**IoE Design Methodology**		08	CO2 and CO4
	2.1	Generic Design Methodology		
	2.2	Application Development Steps		
	2.3	Components of IoE: Processors, Interfacing of Sensors and Actuators		
	2.4	RTOS (Real Time Operating System)		
3	**Programming for IoE Systems**		10	CO2
	3.1	Basic Arduino Programming		
	3.2	Extended Arduino Libraries		
	3.3	Connectivity and Communication in IoE (Protocols and Logical Design)		
	3.4	System Development using Raspberry PI		
4	**IoE Analytics**		04	CO3
	4.1	Business Process in IoE		
	4.2	IoE Analytics with cloud		
	4.3	Edge analytics		
5	**Applications of IoE in**		04	CO4
	7.1	Agriculture		
	7.2	Health Care		
	7.3	Industrial Application		
	7.4	Smart Cities		
	7.5	Environment, Energy		
		Total	30	

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Arshdeep Bhaga and Vijay Madisetti	"Internet of Things (A Hands-on-Approach)",University Press	Tata McGraw-Hill ,India	4th edition ,2015
2	Hakima Chaouchi	"The Internet of Things (Connecting objects to the web)"	Wiley publication	1st edition,2014
3	Raj Kamal	INTERNET OF THINGS (IOT): Architecture and Design Principles	Tata McGraw-Hill ,India	2nd Edition, 14 June 2022
4	Marina Ruggieri, Homayoun Nikookar,	"Internet of Things - From Research and Innovation to Market"	River Publisher,	1st edition,2014
5	Ahriram K Vasudevan,Abhishek Nagarajan,RMD Sundaram	Internet of Things	Wiley Publication	1st edition ,2019
6	Adrian Mcewen, Hakim Cassimally	Designing the Internet of Things	Wiley Publication	1st edition,2014
6	https://iot.eclipse.org/, https://www.mathworks.com/products/thingspeak.html last accessed on August 6, 2025			
7	https://nptel.ac.in/courses/106105166 last accessed on August 6, 2025			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL756	Internet of Everything- Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course
216U01E761	Blockchain Technology : Platforms and Applications

Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Basic of Blockchain
- Networking Concepts,
- Object Oriented Programming Skills,
- Cryptography
- Network Security Concepts.

Course Objectives:

The objective of this course is to provide conceptual understanding of how block chain technology can be used to innovate and improve business processes. The course covers the technological underpinning of block Chain operations in both theoretical and practical implementation of Solutions using block Chain technology.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Develop applications in Solidity language & Multiple Technology-based developments.

CO2. Develop block chain based solutions and write smart contract using Hyperledger Fabric and Ethereum frameworks.

CO3. Illustrate in-depth understanding of Blockchain, Smart Contracts & it's working

CO4. Integrate ideas from various domains and implement them using block chain technology in different perspectives.

CO5. Use Block chain Technology for practical applications

Detailed Curriculum

Module No.	Unit No.	Details	Hrs.	CO
1	**Smart Contract Development and Deployment**		**10**	**CO2 CO3**
	1.1	Introduction to Smart Contracts, Types of Smart Contracts, Structure of a Smart Contract, Smart Contract Approaches, Limitations of Smart Contracts.		
	1.2	**Programming for Blockchain:** Introduction to Programming: Solidity Programming – Basics, functions, Visibility and Activity Qualifiers, Address and Address Payable, Bytes and Enums, Arrays-Fixed and Dynamic Arrays, Special Arrays-Bytes and strings, Struct, Mapping, Inheritance, Error handling.		
2	**Go Ethereum**		**6**	
	2.1	Introduction to Go Ethereum (Geth), Geth Installation and Geth CLI, Setting up a Private Ethereum Blockchain. **Introduction to Truffle**: Smart Contract deployment on a Private Blockchain. Introduction to Ganache.		**CO1**
	2.2	Development Frameworks: Truffle (Development, Migration, Testing), Hardhat (Deployment, Debugging, Plugins), Ganache (Local Blockchain Simulation), MetaMask (Wallet & dApp Integration).		
3	**Enterprises Blockchain**		**8**	
	3.1	**Hyperledger Fabric Chaincode:** Chaincode, Gradle, Chaincode Development Chaincode Package, Install, Approve.		**CO2 CO5**
	3.2	**Multichain:** Introduction to Multichain, Multichain Installation, Create a Multichain Instance, Multichain Assets, Multichain Streams, Multichain Consensus, Multichain API.		
4	**Use Cases & Applications**		**6**	
	4.1	1. Setup Hyperledger Fabric. 2. Set up Multichain in the Local Machine. 3. Create a Private Multichain Blockchain.		**CO4**
	4.2	Layer-2 Solutions and Scalability: Sidechains, State Channels, Plasma, Rollups (Optimistic & ZK-Rollups), Cross-chain Interoperability (Polkadot, Cosmos).		
		Total	**30**	

Recommended Books:

Sr. No.	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
4.	Andreas M. Antonopoulos	Mastering the Bitcoin: Programming the Open Blockchain	O' Reilly	2nd Edition, 2017
5.	Melanie Swan	BlockChian	O'Reilly	2015
6.	Nitin Gaur, Luc Desrosiers, Petr Novotny, Venkatraman Ramakrishna	Hands-On Blockchain with Hyperledger: Building decentralized applications with Hyperledger Fabric and Composer	Packt	Kindle Edition, 2018
4.	*Stephen Fleming business ecosystems*	*Blockchain Technology: Introduction to Blockchain Technology and its impact on*	Stephen Fleming	2017
5.	*Zeeshan-ul- hassan Usmani*	*Introduction to lockchain with Case Studies*	Guhftgu Publication	2018

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL761	Blockchain Technology : Platforms and Applications Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Game Programming			
216U01E762	Game Programming			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Computer Graphics, Mathematics, Basic Physics, Data Structure and Algorithms

Course Objectives:

1. Understand the concepts of Game design and development
2. Learn the processes, mechanics and issues in Game Design.
3. Be exposed to the Core architectures of Game Programming.
4. Know about Game programming platforms, frame works and engines.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand and choose appropriate game architecture

CO2. Design a game using appropriate development environment

CO3. Implement a game with detailed components

CO4. Optimize, debug and publish an ethical games

Detailed Curriculum

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to game programing**		04	CO1
	1.1	Game Architecture,Applying the Game Architecture,Application Layer,Game Logic		
	1.2	Game View for the Human Player,Game Views for AI Agents,Networked Game Architecture,Design Philosophy of DirectX		
2	**Hardcore Game Programming**		6	CO2
	2.1	Introduction to Game Design Introduction to Physics Modeling		
	2.2	Data Structures and Algorithms for Game Programming		
	2.3	The Mathematical Side of Games:Coordinate systems (2D),Vectors and vector math (dot, cross product),Transformations (translation, rotation, scaling),Matrices, projections,Basic trigonometry and its use in game development		
	2.4	Introduction to Physics Modeling:Rigid body dynamics,Gravity, friction, acceleration,Collision response and triggers,Using physics engines (Unity Physics, Box2D)		
3	**Core Game Architecture & Design**		05	CO3
	3.1	Object-oriented game design principles,Game objects, components, prefabs		
	3.2	Game state management (menus, pause, resume),Scripting languages: C# (Unity), C++ (Unreal), or GDScript (Godot)		
4	**2D Game Programming**		8	CO3
	4.1	Sprites and animation,Input handling (keyboard, mouse, touch)		
	4.2	Collision detection (AABB, circle),Score, HUD, sound, and game UI		
	4.3	Simple player movement: Gravity and jumping,Sprites, layering, sprite sheet animation, Parallax background and particle systems		
	4.4	Self-reading: Build a 2D game (platformer, shooter, etc.)		
5	**Game Development Tools, debugging & Deployment**		7	CO3 CO4
	5.1	Asset management,Building and testing on multiple platforms (PC, Android, Web)		
	5.2	Optimization and Debugging The Art of Handling Failure Debugging Basics Graphics and Shader Debugging Debugging Techniques ,Game optimization and performance tuning		
	5.3	Introduction to monetization, publishing, and licensing		
		Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Mike "MrMike" McShaffry and David "Rez" Graham	*Game Coding Complete*	Cengage Learning. USA	Fourth Edition 2013
2	André LaMoth	*Game Programming All in One*	Premier Press, Inc, USA	Second Edition , 2002

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL762	Game Programming - laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E763	High Performance Computing			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Computer Organization and Architectures, Data structures, Analysis of algorithms

Course Objectives:

1. To introduce the students with various Parallel architectures that support Design and Development of parallel algorithms/programs so as to achieve super linear speedup.
2. The course introduces parallel programming paradigms using tools like MPICH, OpenMP and multicore program approaches using CUDA platform.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Explain and classify different architectures for parallel systems.
CO2. To identify and analyse data, control, and loop-carried dependencies in algorithms to determine their impact on parallelisation.
CO3. Use different Parallel Programming models and Parallel Programming tools for implementing parallel algorithms.
CO4. Describe CUDA Architecture with their memory structure.
CO5. Demonstrate parallel programming for different real-world applications using CUDA.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to High Performance Computing**		06	CO1
	1.1	Parallel processing Concepts: Levels of parallelism (instruction, transaction, task, process, thread, memory, and function), Pipelining, data parallelism, control parallelism.		
	1.2	Parallel Architectures: Shared memory multiprocessors (UMA Uniform), Distributed memory multiprocessors (NUMA Non Uniform memory Access).		
	1.3	Cluster computing, Multicore Systems, Dataflow Models, Systolic processors		
	1.3	Self-learning: Applications of High-Performance Computing: e.g Climate modelling, Healthcare etc.		
2	**Data Dependency Analysis and Parallel Algorithms**		07	CO2
	2.1	Data dependency analysis		
	2.2	Introduction to parallel algorithms: Preliminaries, Decomposition Techniques, characteristics of tasks & interactions, mapping techniques for load balancing, methods for containing interaction overheads. Parallel algorithm models.		
	2.3	Parallel Algorithms: Parallel reduction, sorting algorithms: bubble / quick sort; Matrix multiplication.		
	2.4	Self-learning: Performance Analysis of Parallel Algorithms.		
3	**Parallel Programming Models and Platforms**		06	CO3 CO1
	3.1	Parallel programming models: shared memory programming model		
	3.2	Parallel programming models: Message passing programming model		
	3.3	Parallel programming Tools/platforms: MPICH, OpenMP, High performance JAVA(Any two)		
4	**CUDA : GPU Parallel Development Environment**		06	CO4
	4.1	Compute Unified Device Architecture (CUDA) Architecture, CUDA programming model, execution model.		
	4.2	Thread organization: Concept of grid, block and thread; thread index generation, warp.		
	4.3	GPU primitives: scan (exclusive or inclusive), scatter, gather, reduce, memory model.		
	4.4	Introduction to global, shared, local memories, usage of cache, texture cache, constant memory.		
	4.5	Self-learning:CUDA structure, API and library (CUDPP, CUBLAS, FFT etc.)		
5	**GPU parallel programming applications**		05	CO5
	5.1	CUDA example programs (Vector dot product, Matrix multiplication (with the usage of tiling and shared memory)		
	5.2	Graph algorithms, molecular dynamics, n-body simulations(any two) using GPU.		
	5.3	Tensor Processing Unit		
	5.4	**Self-reading:** Tensor flow		

Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Ananth Grama, George Karypis, Vipin Kumar, Anshul Gupta	*"Introduction to Parallel Computing"*	Pearson	Edition 2, 2015
2	Fayez Gebali	*Algorithms and Parallel Computing*	Wiley Publication	Edition 1, 2011
3	M. J. Quinn	*Parallel Programming in C with MPI and OpenMP*	McGraw Hill	Edition 1, 2004
4	Edward Kandrot and Jason Sanders	*CUDA by Example: AN Introduction to General Purpose GPU Programming*	Addison-Wesley Professional	Edition 1, 2010
5	Duane Storti, Mete Yurtoglu	*Cuda for Engineers: An Introduction to High Performance Parallel Computing*	Pearson Education India	Edition 1, 2016
6	https://www.tensorflow.org/learn			

*In addition to printed books, faculty can suggest (authentic) URLs or e-books, e-contents etc.

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL763	High Performance Computing-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E764	Internet of Things Security			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Microcontroller and embedded system, IOT, Information Security.

Course Objectives:

1. Understand the fundamentals of Internet of Things (IoT), its architecture, layered models
2. Identify and analyze security threats, vulnerabilities, and risks associated with IoT devices, networks, and data.
3. Apply cryptographic techniques and authentication mechanisms in IoT environments.
4. Explore IoT security frameworks and standards.
5. Evaluate emerging technologies such as AI/ML, Blockchain, and Zero Trust models for strengthening future IoT security applications across domains.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Understand IoT architecture and potential security vulnerabilities.

CO2. Identify and assess security threats, vulnerabilities, and risks in IoT devices and networks.

CO3. Apply lightweight cryptographic and authentication mechanisms for securing IoT communications.

CO4. Design secure IoT architectures by integrating standard frameworks, policies, and best practices.

CO5. Evaluate and propose innovative security solutions using AI, Blockchain, and hardware-based security for future IoT ecosystems.

Module No.	Unit No.	Contents	No. of Hrs.	CO
1	**Introduction to IoT**		02	CO1
	1.1	IoT fundamentals, IoT devices, components of IoT & its architecture and layered model, communication models and protocols.		
	1.2	Growth and Impact of IoT, Implementation challenges.		
	1.3	IoT applications across domains (smart home, healthcare, industrial, etc.)		
2	**Security Fundamentals for IoT**		07	CO1 CO2
	2.1	Attacks on IoT devices: TVR (Threat, Vulnerability, Risks), Attack vectors, Logical Attack, Invasive, Semi-Invasive and Non-Invasive Attacks.		
	2.2	Threat modeling in IoT, Common IoT attacks: Physical attacks (tampering, side-channel attacks), Network attacks (DoS/DDoS, routing attacks, spoofing, MITM), Software attacks (malware, firmware vulnerabilities), Data-Centric Attacks: Data theft, Unauthorized access to cloud platforms, Privacy breaches, Privacy challenges in IoT systems.		
	2.3	Attack Models – Attacks to Sensors in IoTs, Attacks to RFIDs in IoTs,Attacks to Network Functions in IoTs,Attacks to Back-end Systems,Security in Front-end Sensors and Equipment, Prevent Unauthorized Access to Sensor Data,M2M Security, RFID Security, Cyber-Physical Object Security, Hardware Security.		
3	**Cryptography & IoT**		07	CO3
	3.1	Cryptography: Lightweight encryption (AES, ECC, ChaCha20), Secure key management for constrained devices.		
	3.2	Authentication & Identity Management: Mutual authentication protocols (DTLS, TLS, Lightweight EAP), Device certificates and public key infrastructure (PKI). Access Control: Role-based access control (RBAC), Attribute-based access control (ABAC), Capability-based models for constrained devices.		
	3.3	Secure Communication: End-to-end encryption, Protocol security (CoAPs, MQTT-SN with TLS/DTLS), Firmware & Patch Management: Secure boot and code signing, Over-the-air (OTA) updates with integrity verification, Intrusion Detection & Monitoring, Anomaly detection for IoT traffic, Edge and cloud-based monitoring systems, Authentication protocols (e.g., OAuth, MQTT authentication, CoAP security).		
4	**Secure IoT Architectures and Standards**		07	CO3
	4.1	IoT security frameworks (NIST, ENISA, ISO/IEC), Secure device bootstrapping and provisioning, Access control models for IoT.		
	4.2	Blockchain for IoT security, Standards and best practices: IEEE, IETF, ITU-T initiatives.		

	4.3	IoT Security Frameworks: NISTIR 8259 (IoT Device Cybersecurity) ENISA IoT Security Guidelines, ISO/IEC 27030 IoT Security and Privacy		
	4.4	Self-reading: Mirai Botnet (2016), Stuxnet, Smart Home Privacy Breaches		
5	IoT Security Future & Applications.		07	CO3 CO4
	5.1	AI/ML-based anomaly detection in IoT networks, Blockchain for decentralized trust and secure device identity, Zero Trust architectures for IoT, Hardware-based security (TPM, PUFs, secure enclaves).		
	5.2	Applications of IoT Security: Smart Homes, Healthcare (IoMT – Internet of Medical Things), Industrial IoT (IIoT) & Smart Manufacturing, Smart Cities, Transportation and Automotive (IoV – Internet of Vehicles), Agriculture (Smart Farming), Retail and Supply Chain, Energy and Smart Grids, Military and Defense, Financial Services (FinTech IoT).		
	5.3	Self-reading: Case-studies pertaining to future evolvement of technologies to support IoT Security applications.		
		Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Fei HU	Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations	CRC Press	1st Edition, 2020
2	Brian Russell, Drew Van Duren	Practical Internet of Things Security: Design a security framework for an Internet connected ecosystem	Packt	2nd Edition, 2018
3	John R. Vacca	Security and Privacy in Internet of Things: Challenges and Solutions	CRC Press	2020
4	Ollie Whitehouse	Security of Things: An Implementers' Guide to Cyber-Security forInternet of Things Devices and Beyond	NCC Group	1st Edition, 2014
5	Rajkumar Buyya, Amir Vahid Dastjerdi	Internet of Things: Principles and Paradigms	Morgan Kaufmann	2016

Course Code	Name of the Course			
216U01EL764	Internet of Things Security-Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Course Code	Name of the Course			
216U01E765	Introduction to Large Language Models			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basic NLP Knowledge, Machine learning & deep learning concepts and python programming.

Course Objectives:

1. Introduce students to the architecture and evolution of large language models and their underlying transformer foundations.
2. Enable understanding of pretraining, fine-tuning, and evaluation techniques for building and optimizing LLMs.
3. Develop skills in prompting and in-context learning for zero-shot and few-shot NLP tasks.
4. Familiarize students with real-world applications of LLMs in domains like QA, summarization, and code generation.
5. Encourage critical reflection on ethical issues such as bias, hallucination, and responsible deployment of LLMs.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to:

CO1: Understand the core architecture and components of transformer-based language models.
CO2: Apply pretraining and fine-tuning techniques to adapt LLMs for downstream NLP tasks.
CO3: Design and experiment with effective prompts for zero-shot, few-shot, and chain-of-thought reasoning.
CO4: Demonstrate the use of LLMs in solving real-world tasks using tools like OpenAI API or HuggingFace.
CO5: Evaluate the ethical, safety, and social implications of deploying large-scale language models.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction and Architecture of LLMs**		06	CO1
	1.1	Evolution from traditional NLP to LLMs, Scaling laws and parameter growth.		
	1.2	Pretraining objectives: causal LM (CLM) vs masked LM (MLM)		
	1.3	Model architectures: Decoder-only (GPT), Encoder-only (BERT), Encoder-Decoder (T5, BART)		
	1.4	Self-learning: Architectural Innovations in Open-Source LLMs: LLAMA, MPT, Falcon		
2	**Pretraining and Fine-tuning Techniques**		06	CO2
	2.1	Data collection and pre-processing at scale, Pretraining: objectives, compute, and optimization, Model Quantization.		
	2.2	Fine-tuning paradigms: Task-specific fine-tuning, Instruction tuning & Adapter-based tuning (LoRA, PEFT)		
	2.3	Transfer learning in LLMs, Evaluation metrics for language models (perplexity, BLEU, ROUGE, accuracy).		
	2.4	#Self-Learning: Data Curation and Dataset Filtering Strategies for Pretraining LLMs		
3	**Prompting and In-Context Learning**		06	CO3
	3.1	Zero-shot, one-shot, and few-shot prompting		
	3.2	Prompt engineering: basics and patterns, Chain-of-thought prompting (CoT)		
	3.3	Retrieval-Augmented Generation (RAG) Architecture – Retriever & Generator– concept only		
	3.4	#Self-Learning: OpenAI Playground, LangChain.		
4	**Applications of LLMs**		06	CO4
	4.1	Text classification, summarization, language translation		
	4.2	Code generation, question answering, Conversational agents and chatbots		
	4.3	Multimodal LLMs (e.g., GPT-4V, Flamingo)		
	4.4	Self-reading: OpenAI, HuggingFace, Cohere, Google PaLM		
5	**Ethics, Safety, and Limitations**		06	CO5
	5.1	Model bias and fairness, Hallucinations and factuality challenges		
	5.2	Privacy and data leakage, Regulation and open-source vs proprietary models.		
	5.3	Responsible deployment and LLM governance		
	5.4	Self-reading: Red Teaming and Alignment Strategies in Large Language Models		
		Total	30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Tanmoy Chakraborty	Introduction to Large Language Models Generative AI for Text	Wiley	1st Edition, Dec 2024
2	Jay Alammar, Maarten Grootendorst	Hands-On Large Language Models: Language Understanding and Generation	O'Reilly	1st Edition, Oct 2024
3	Sebastian Raschka	Build a Large Language Model from Scratch	Manning Publications	1st Edition, Oct 2024
4	Chip Huyen	AI Engineering: Building Applications with Foundation Models	O'Reilly	1st Edition, Dec 2024
5	Valentina Alto	Building LLM Powered Applications: Create intelligent apps and agents with large language models	Packt Publications	1st Edition, May 2024
6	Weblinks: 1.Natural Language Processing with Transformers, Revised Edition [Book] 2. Hands-On Large Language Models [Book] 3. The Illustrated Transformer – Jay Alammar – Visualizing machine learning one concept at a time. 4. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding - ACL Anthology 5. language_understanding_paper.pdf 6. [1910.10683] Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer 7. Safety & responsibility | OpenAI 8. Parameter-Efficient Fine-Tuning using 🤗 PEFT 9. [2302.13971] LLaMA: Open and Efficient Foundation Language Models 10. [2203.02155] Training language models to follow instructions with human feedback 11. Aligning language models to follow instructions | OpenAI 12. [2302.13971] LLaMA: Open and Efficient Foundation Language Models			
7	NPTEL course: 1. Introduction to Large Language Models (LLMs) - Course 2. Introduction - Hugging Face LLM Course			
8	Coursera Course: 1. Best Large Language Models (LLMs) Courses & Certificates Online [2025] | Coursera Learn Online 2. Generative AI with LLMs by AWS & DeepLearning.AI | Coursera			

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL765	Introduction to Large Language Models-Lab			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01

Evaluation Scheme	Marks				
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

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Course Code	Name of the Course			
216U01E766	MLOPS			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	02	--	--	02
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basic knowledge of programming (preferably Python), understanding of software development lifecycle, familiarity with machine learning fundamentals, and working knowledge of Linux command-line operations.

Course Objectives:

1: To introduce the foundational concepts and lifecycle of DevOps and MLOps in modern software and ML systems.

2: To equip learners with hands-on skills in version control, CI/CD pipelines, and infrastructure automation.

3: To enable the design and implementation of reproducible and automated ML workflows using industry-standard MLOps tools.

4: To familiarize learners with containerization, orchestration, and cloud deployment for scalable application and model delivery.

5: To develop the ability to monitor, secure, and manage production-grade ML systems through observability and governance practices.

Course Outcomes (CO):

CO1 Apply DevOps tools and practices to automate software development, testing, and deployment workflows

CO2 Design and implement MLOps pipelines for scalable machine learning model development and deployment.

CO3 Deploy DevOps and MLOps solutions on cloud platforms using infrastructure-as-code and managed ML services.

CO4 Monitor and secure ML models in production using observability tools and DevSecOps principles

CO5 Execute an end-to-end real-world project integrating DevOps and MLOps for continuous delivery of intelligent applications.

Module No.	Unit No.	Details	Hrs.	CO
1	**Introduction to DevOps & MLOps**		4	1
	1.1	DevOps vs MLOps: Key Differences & Overlaps Importance of Automation in Software & ML Workflows		
	1.2	DevOps & MLOps Lifecycle, CI/CD for Applications and ML Models		
2	**Version Control & Collaboration**		6	1
	2.1	The need for Source Code, Overview of Version Control systems Setting up a basic, Git server, Hosted Git Server		
	2.2	Branching and merging in Git, Git Server implementation, Working with Remote repository		
3	**Continuous Integration and Delivery**		5	2
	3.1	Jenkins Architecture, Plugin Management in Jenkins, Jenkins Security Management, Notification in Jenkins, Jenkins Master-slave architecture, Jenkins Delivery Pipeline, Jenkins Declarative pipeline		
	3.2	ML-specific CI/CD using Jenkins, GitHub Actions Model validation, testing, and approvalML pipelines: Kubeflow Pipelines, Airflow, ZenML		
4	**Containerization and Orchestration with Kubernetes.**		5	3
	4.1	Docker: Image creation, networking, multi-stage builds, Dockerizing ML/AI applications, Docker Compose for multi-container services, Conda and virtualenv for reproducible environments		
	4.2	Kubernetes core components (Pods, Deployments, Services), Helm charts for packaging Kubernetes apps, Model deployment on K8s using Seldon Core / KFServing, Hands-on: ML app deployment on Minikube / EKS		
5	**Infrastructure as Code (IaC) and Configuration Management**		5	4
	5.1	Terraform basics and provisioning cloud infra, Ansible for configuration management, AWS CloudFormation (optional), Creating reusable infrastructure templates.		
6	**Monitoring using Prometheus and Grafana**		5	5
	6.1	Monitoring using Prometheus and Grafana, Introduction to Prometheus and Grafana, Prometheus and Grafana Setup, Monitoring using Prometheus, Dashboard Visualization using Grafana, Creating a Dashboard to monitor the Pipeline.		

	6.2	**Model Serving and API Deployment:** FastAPI/Flask model serving,Model versioning and rollback strategies, REST vs gRPC for inference Deployment with BentoML or TorchServe		
Total			30	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Sricharan Vadapalli	DevOps: Continuous Delivery, Integration, and Deployment with DevOps: Dive	Packt	1st Edition, 2018
2	Emily Freeman	DevOps for Dummies	Wiley	1st Edition, 2022
3	Kelsey Hightower, Brendan Burns, Joe Beda	Kubernetes: Up & Running	O'Reilly Media	3rd Edition, 2022
4	Nigel Poulton	The Kubernetes Book	Kindle	2024 Edition

Laboratory/Tutorial

Course Code	Name of the Course			
216U01EL766	MLOPS Laboratory			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	--	02	--	02
Credits Assigned	--	01	--	01
Evaluation Scheme	Marks			
	LAB/TUT CA*	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

***LAB/TUT CA is an evaluation carried out during the said laboratory/tutorial throughout the semester on a continuous basis. In case of Laboratory, it can be a combination of laboratory experiments performed (at least 8-10), written record of experiments (Journal), Viva/On-screen test and/or Quiz, programming assignments (wherever applicable) and practical examination (if any) conducted during the semester. In case of tutorial, it can be a combination of graded assignments, group/individual activities such as presentations, group discussion, report writing etc. (as applicable). For PBL course, the Lab/Tut CA marks are assigned for project evaluation, however, some marks may be reserved for laboratory experiments performed (in any) and tutorials (if any).**

Open Electives Technical 2

Course Code	Name of the Course			
216U01O701	Genetic Algorithm and Applications			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course prerequisites (if any):
Data Structures, Algorithms, Basic Python Programming, Probability & Statistics

Course Objectives
The objective of this course is to introduce optimization techniques and application of genetic algorithms. The course also familiarizes students with the concepts of various operators and their implementation. Course mainly focuses on applying the genetic algorithm for a real life application.

Course Outcomes
At the end of successful completion of the course the student will be able to

CO1 Describe the concept of optimization

CO2 Understand the concept of genetic algorithms

CO3 Implement the component of GA to solve the problems.

CO4 Illustrate the application of GA in various domains.

Module No.	Unit No.	Details	Hrs.	CO
1	**Introduction To Optimization**		03	CO1
	1.1	Finding the Best Solution - What Is Optimization? , Root Finding versus Optimization, Categories of Optimization		
	1.2	Minimum-Seeking Algorithms - Exhaustive Search, Analytical Optimization, Nelder-Mead Downhill Simplex Method, Optimization Based on Line Minimization,		
	1.3	Nature-Inspired Algorithms: An Overview Evolution, Swarm Intelligence, and Natural Selection Principles		
2	**Introduction to Genetic Algorithms**		08	CO1 , CO2
	2.1	What are genetic algorithms? Analogy of GA, Robustness of Traditional Optimization and Search methods – Goals of optimization-GA versus Traditional methods		
	2.2	Mathematical foundations: The fundamental theorem - Schema processing at work. – The 2-armed & k-armed Bandit problem. – The building Block Hypothesis. – Minimal deceptive problem		
3	**The Basic of Genetic Algorithm**		12	CO2
	3.1	Basic flow of a genetic algorithm, Selection Methods – Roulette wheel selection, stochastic universal sampling, rank-based selection, fitness scaling, tournament selection		
	3.2	Crossover Methods – Single, Two and k – point crossover, uniform crossover, crossover for ordered list		
	3.3	Mutation Methods – Flip bit mutation, swap mutation, inversion mutation, scramble mutation		
	3.4	Real-coded genetic algorithms – blend crossover and simulated binary crossover. Understanding elitism, Constraint Handling Methods in GA		
4	**Solving Problems with Genetic Algorithms**		12	CO2 ,CO 3
	3.1	Using the DEAP Framework – Introduction to DEAP, Using the creator module (fitness class), Using the Toolbox class (genetic operators) , OneMax Problem		
	3.2	Search Problems and Combinatorial Optimization –Knapsack and TSP		
	4.1	Constraint Satisfaction in search problems, Solving scheduling and N-queens problems		
	4.2	Optimizing Continuous Functions – chromosomes and genetic operators for real numbers, Using DEAP with continuous functions		
5	**Applications of Genetic Algorithms**		10	CO3 ,CO 4
	5.1	Enhancing Machine Learning Models Using Feature Selection		
	5.2	Hyperparameter Tuning of Machine Learning Models		
		Total	45	

Recommended books

Sr. No	Name/s of Author/s	Title of Book	Name of Publisher with country	Edition and Year of Publication
1	Eyal Wirsansky	Hands-On Genetic Algorithms with Python	Packt Publishing ISBN: 9781838557744	Released January 2020
2	David E. Goldberg	*Genetic Algorithms in Search, Optimization and Machine Learning*	Pearson Education	2021
3	S.Rajesekaran, G.A.Viijayalakshmi Pai	*Neural Networks, Fuzzy Logic and Genetic Algorithms*	Pearson Education	2003
4	Banzhaf, Nordin, Keller and Francone, Morgan-Kaufmann	*Genetic Programming: An Introduction*	Morgan Kaufmann	February 1998
5	Riccardo Poli, Willian B. Langdon, Nicholas Freitag McPhee	*A Field Guide to Genetic Programming*	Lulu.com	March 2008
6	Clinton Sheppard	*Genetic Algorithms With Python*	Createspace Independent Pub	April 2016
7	Introduction to Genetic Algorithms	Melanie Mitchell	MIT Press	2023
8	John Hearty	Applied Evolutionary Algorithms in Python	O'Reilly	2021

Course Code	Name of the Course			
216U01O702	Reinforcement Learning			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basics of AI

Course Objectives:

1. Comprehension of fundamentals of reinforcement learning
2. Application of various RL methods for
3. Analysis using temporal difference and deep learning methods
4. Evaluation of various methods of RL for application development

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Interpret fundamentals of Reinforcement learning methods
CO2. Apply various dynamic programming and Monte-Carlo methods
CO3. Apply different temporal difference learning policies
CO4. Use n-step bootstrapping policies and apply planning for different models
CO5. Understand future of reinforcement learning

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction to Reinforcement Learning**		08	CO1
	1.1	Elements of Reinforcement Learning, Limitations and scope		
	1.2	Markov Decision Processes to solve real-world problems, Goals and rewards, Policies and value functions		
	1.3	Bellman equations for Markov Decision Processes		
		Self-learning:History of RL		
2	**Dynamic Programming and Monte Carlo Methods**		10	CO2
	2.1	Iterative policy evaluation, Policy improvement, Policy iteration and value iteration		
	2.2	Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency of Dynamic Programming		
	2.3	Classic Monte Carlo prediction and control methods, Greedy and epsilon-greedy policies, Exploration-Exploitation Dilemma and its solutions		
	2.4	#Self-Learning: A/B testing and bandit algorithm		
3	**Temporal Difference Learning**		10	CO3
	3.1	TD Prediction, Advantages of TD Prediction Methods		
	3.2	Sarsa: On-policy TD Control		
	3.3	Q-learning: Off-policy TD Control		
	3.4	#Self-Learning:Deep learning,Generative models		
4	**n-step Bootstrapping**		09	CO4
	4.1	n-step TD Prediction, n-step Sarsa, n-step Off policy		
	4.2	Off policy Learning Without Importance Sampling: The n-step Tree Backup Algorithm		
	4.3	Models and Planning, Dyna: Integrated Planning, Acting, and Learning		
	4.4	**Self-reading: n-step TD learning eligibility trace proofs**		
5	Deep Q-Learning and Reinforcement Learning Lifecycle		08	CO5
	5.1	Deep Q-learning: Experience Replay, Implementing DQ N with examples.		
	5.2	RL project life cycle, problem definition for RL projects		
	5.3	Future of reinforcement learning, RL market opportunities, Future RL and research directions		
	5.4	**Self-reading: RL debugging**		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Richard S. Sutton and Andrew G. Barto	*Reinforcement Learning: An Introduction*	The MIT Press	Second Edition, 2018
2	Stuart Russell and Peter Norvig	*Artificial Intelligence: A Modern Approach*	Pearson	3 rd edition, 2010
3	Robert Tibshirani, and Jerome Friedman	*The Elements of Statistical Learning, Trevor Hastie*	Springer	2nd edition, 2009
4	M. Gopal	*Applied Machine Learning*	Mc-Graw Hill Education India Pvt. Ltd.	Print edition: ISBN-13: 978-93-5316-025-8,
5	Chris Bishop	*Pattern recognition and machine learning (PRML)*	Springer	ISBN-13: 978-0387-31073-2, 2006
6	Phil Winder	Reinforcement Learning	O'reilly	ISBN-978-93–8588-950-9

Course Code	Name of the Course			
216U01O703	Web Analytics and Web Intelligence			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	02	--	--	02

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	150

Course pre-requisites: Basics of Web Programming

Course Objectives:

1. Web Analytics refers to the measurement, collection, analysis, and reporting of Internet data with the main goal of understanding user behaviour and optimizing overall web usage.
2. It serves as a tool to measure website traffic while also supporting business and market research to improve decision-making and online performance.
3. The course covers the definition and categories of Web Analytics, highlights popular tools like Google Analytics, and explains the usage of the EDDIE Tool for deeper analysis and reporting.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to:

CO1: Understand key concepts, terms, and parameters of web analytics and their impact on business insights.

CO2: Understand the concepts and terminologies related to web intelligence.

CO3: Explore the use of tools and techniques used in web analytics.

CO4: Develop experience on building websites, analysing web data insights and conversions.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Web Analytics Foundations & Customer Insights**		10	CO1
	1.1	Basics & Traditional Data Collection: Foundational concepts of analytics. Traditional ways of gathering data: clickstream data, weblogs, and beacons. Role of JavaScript tags and packet sniffing in capturing user behavior.		
	1.2	Expectations & Advanced Data Sources:Expected outcomes from analytics , advanced data types: outcomes data, competitive data, and search engine data.		
	1.3	Customer Centricity & Site Visits: Customer behavior : site visits, surveys, questionnaires, website surveys, and post-visit feedback.		
	1.4	Strategic Insights & Benefits: Benefits of surveys ,critical components of a successful strategy to drive customer-focused improvements.		
	1.5	#Self-Learning:		
2	**Qualitative Web Intelligence**		08	CO2
	2.1	Customer-Centric Design: Journey Mapping and setting Experience Goals to enhance user interactions and satisfaction.		
	2.2	User Feedback Tools: NPS, On-page feedback, and Polls to gather insights on user experience and preferences.		
	2.3	Analytics Tools Overview: Hotjar, Microsoft Clarity, and Smartlook, including features like Heatmaps, Scroll Maps, and Session Recordings for visual behavior analysis.		
	2.4	Using Surveys and Qualitative Data: Identifying drop-off points, friction areas, and crafting actionable insights to improve website performance and user experience.		
	2.5	#Self-Learning:		
3	**Metrics, Dimensions, and Engagement Analytics**		09	CO3
	3.1	Web Metrics: Pageviews, Engagement Rate, Time on Site, Scroll Depth, and Events to evaluate user interactions and website performance.		
	3.2	Dimensions: Device, Geography, and Traffic Source for segmenting and analysing user data effectively.		
	3.3	Navigation & Content Reports: Landing Pages, Exit Pages, Flow Reports, and Campaign Parameters like UTM tracking to measure campaign performance.		

	3.4	Search & Cross-Channel Analytics: Internal Search, SEO basics, PPC metrics, and performance insights across Email, Social, Referral, and Paid channels.		
	3.5	#Self-Learning:		
4	**Google Analytics 4 (GA4) in Practice**		09	CO3
	4.1	GA4's Event-Driven Model focuses on Events, Parameters, and Conversions to provide user-centric tracking, along with Real-Time Reporting, Debugging, Data Streams, and an introduction to BigQuery Export for advanced analytics.		
	4.2	Audience Building and Custom Reports enable segmentation and tailored insights to better understand user behavior and performance metrics.		
	4.3	Google Tag Manager (Introduction) allows Event setup without code and provides access to Demographics and Tech Insights for deeper analysis.		
	4.4	**Self-reading:**		
5	**Clustering and Mining Social Network and graphs**		08	CO3 CO4
	5.1	Creating Custom Funnels and Conversion Paths in GA4, along with Ecommerce and Event Tracking Basics, to analyze user journeys and key interactions.		
	5.2	Attribution Models in GA4: Understanding Data-driven, Last Click, Position-based, and Multi-touch Attribution and their importance in measuring marketing effectiveness.		
	5.3	Linking GA4 with Google Ads and Search Console and building Real-Time Dashboards using Looker Studio (formerly Data Studio) for integrated insights and visualization.		
	5.4	**Self-reading:**		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Avinash Kaushik	Web Analytics 2.0: The Art of Online Accountability and Science Of Customer Centricity	Sybex	1st edition, 2009
2	Michael Beasley	Practical Web Analytics for User Experience: How Analytics can help you Understand your Users	Morgan Kaufmann	2013
3	Magy Seif El-Nasr, Anders Drachen, Alessandro Canossa, eds.	Game Analytics: Maximizing the Value of Player Data	Springer	2013
4	Bing Liu	Web Data Mining: Exploring Hyperlinks, Content, and Usage Data	Springer	2nd Edition, 2011
5	Justin Cutroni	Google Analytics	O'Reilly	2010
6	Eric Fettman, Shiraz Asif, Feras Alhlou	Google Analytics Breakthrough	John Wiley & Sons	2016

Course Code	Name of the Course			
216U01O704	Massive Graph Analysis			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites: Basic of Graphs and Visualisation

Course Objectives:

1. To explore basic concepts of graph structured data and its Algorithms.

2. To learn different ways of modelling to store, retrieve and analyse the graph Structured data.

3. To understand the applications of graph analytic techniques and advanced concepts in graph analytics

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. To explore the concept of Graphs and To learn modelling and storing graph-structured data and related algorithms

CO2. To retrieve and analyse graph-structured data.

CO3. To understand the advanced concepts in graph analytic techniques.

CO4. To understand applications of graph analytics.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Introduction - Large Scale Graphs**		07	CO1
	1.1	Support for Graph Processing, Mapping Graph Algorithms to Architectures, applications of large Characteristics of large scale graph, Complex data sources, Social Networks, Simulations, Bioinformatics; Categories- Social, Endorsement, Location, Co-occurrence graphs; Graph Data structures, Parallel, Multicore, & Multithreaded Architectural scale graphs.		
2	**Large-scale Graph Analysis**		12	CO2
	2.1	Parallel Prefix & List Ranking, Link Analysis, Page Ranking Algorithms		
	2.2	Parallel BFS, Spanning Tree, Connected Components, Minimum Spanning Tree Matroid Algorithm		
	2.3	Social Networking Algorithms, Parallel Betweeness Centrality		
3	**Dynamic Parallel Algorithms**		10	CO2
	3.1	Streaming Data Analysis -Data Structures for Streaming Data - Tracking Clustering Coefficients		
	3.2	Tracking Connected Components - Anomaly Detection, Massive-Graphs in Computational Biology, Genome Assembly.		
4	**Distributed Computation for Massive Data Sets**		08	CO3
	4.1	Spectral, Modularity-based Clustering, Random Walks; Large Graph Representation and Implementation- V-Graph Representation, Map Reduce ,Surfer, Graph Lab.		
5	**Advanced Massive Graph Analysis topics**		08	CO4
	5.1	Power Law Distribution, Game-Theoretic Approach, Rank Aggregation and Voting Theory, Recommendation Systems, Social network analysis: case study -Facebook,LinkedIn, Google+, and Twitter.		
	5.4	**Self-reading:**		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Matthew O. Jackson	Social and Economic Networks	Princeton University Press	2010
2	Stanley Wasserman, Katherine Faust	Social Network Analysis: Methods and Applications	Cambridge University Press	1995
3	Tanja Falkowski,	Community Analysis in Dynamic Social Networks	University Magdeburg	2009
4	Ladislav Novak, Alan Gibbons	Hybrid Graph Theory and Network Analysis	Cambridge Tracts in Theoretical Computer Science	2009
5	Eric D. Kolaczyk	Statistical Analysis of Network Data Methods and Models	Springer Series in Statistics	2009

Course Code	Name of the Course
216U01O705	**Audio Signal Processing**

Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	–	50

Course pre-requisites:
Linear algebra, Probability and statistics, Multivariate calculus, Algorithms and Programming language such as Python or C++

Course Objectives:
1. Comprehension of fundamentals of audio signal processing
2. Application of signal transforms for audio signal analysis
3. Analysis of sound/audio signal using open-source programming packages and tools
4. Synthesis of audio/sound signal for statistical evaluation
5. Creation of audio/music applications

Course Outcomes (CO):
At the end of the successful completion of the course, the student will be able to
CO1. Understand the principles of audio digitization, including sampling, quantization, and the Nyquist theorem.
CO2. Apply the Fast Fourier Transform (FFT) to compute and interpret the frequency spectrum of an audio signal.
CO3. Analyze a signal's time-frequency content using the STFT and compare it with other transforms.
CO4. Evaluate speech production models like LPC for their ability to estimate vocal tract characteristics.
CO5. Design an audio classification pipeline by selecting appropriate features and machine learning models.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Foundations of Digital Audio & Signals**		08	CO1
	1.1	Signal Fundamentals: Periodicity, aperiodicity, impulse response, convolution.		
	1.2	Digital Representation: **Nyquist-Shannon sampling theorem**, aliasing, quantization, and dithering.		
	1.3	Mathematical Preliminaries: The **Z-transform** and its relation to digital filters (FIR, IIR).		
	1.5	Self-Learning Topic: Study of open source tools for audio processing (Audacity and Praat for visualizing waveforms and spectrograms), Introduction to Python tools sms-tool package		
2	**Frequency Domain Analysis**		10	CO2
	2.1	Discrete Fourier Transform, Inverse-DFT, DFT properties: linearity, shift, symmetry, convolution; energy conservation and decibels; phase unwrapping; zero padding		
	2.2	Fast Fourier Transform and zero-phase windowing, analysis/synthesis, analysis of simple periodic signals and of complex sounds		
	2.3	The Convolution Theorem and its application in fast filtering.		
	2.4	Spectral Leakage and Resolution Trade-offs, The Discrete Cosine Transform (DCT) and its use in compression.		
	2.5	Self-Learning Topic: Sound analysis using DFT: Spectrum of a sound fragment using Python and DFT model functions implemented using sms-tools package		
3	**Time-Frequency Representations**		10	CO3
	3.1	Introduction to STFT, equation and analysis window FFT size and hop size time-frequency compromise, Inverse STFT (The overlap-add method and the Griffin-Lim algorithm)		
	3.2	Introduction to the Wavelet Transform (CWT/DWT) for transient analysis and the Constant-Q Transform (CQT) for musical scales.		
	3.2	Sound/speech production and acoustic phonetics, speech perception, Speech analysis: time and frequency domain techniques for pitch and format estimation, LPC analysis		
	3.3	Introduction to the statistical approach for automatic speech recognition (ASR)		
4	**Models of Speech & Sound Production**		08	CO4
	4.1	The Source-Filter Model of speech production.		
	4.2	Advanced Pitch Detection: Autocorrelation, AMDF, and cepstral methods.		

	4.3	Sound synthesis: articulatory, formant, and LPC synthesis Linear Predictive Coding (LPC).		
	4.4	Introduction to statistical modeling for speech with Hidden Markov Models (HMMs).		
	4.5	4.5 Synthesis: From LPC/formant synthesis to modern statistical parametric and neural vocoder concepts (e.g., WaveNet).		
5	**Feature Engineering & Applications**		09	CO5
	5.1	Perceptual Audio Features: The Mel scale, Mel-Frequency Cepstral Coefficients (MFCCs) and their derivatives.		
	5.2	Advanced Features: Chroma, Spectral Contrast.		
	5.3	Machine Learning for Audio: Using features for classification, clustering, and regression.		
	5.4	Application Deep Dives: Audio Source Separation (e.g., using NMF), Acoustic Scene Classification, and modern ASR pipelines (CNNs on spectrograms, RNNs for sequences).		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Dan Jurafsky and James H. Martin	*Speech and Language Processing*	Stanford University	3rd Edition / 2025
2	Alexander Lerch	*An Introduction to Audio Content Analysis*	Wiley-IEEE Press	2nd Edition / 2020
3	José Unpingco	Python for Signal Processing	Springer	2nd Edition / 2021
4	Udo Zölzer	*DAFX: Digital Audio Effects*	Wiley	2nd Edition/2022
5	Alan V. Oppenheim, Ronald W. Schafer	Discrete-Time Signal Processing	Pearson	3rd Edition / 2009
6	https://www.coursera.org/learn/audio-signal-processing			
7	https://nptel.ac.in/courses/117105145			

Course Code	Name of the Course			
216U01O706	Applied Machine Learning using TensorFlow			
Teaching Scheme (Hrs./Week)	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03

Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30		50

Course pre-requisites: Basic Python Programming, Linear Algebra and Probability

Course Objectives:

1. Understand foundational concepts of machine learning and neural networks, and the role of TensorFlow in ML development.
2. Apply TensorFlow and Keras to build, train, and evaluate various machine learning and deep learning models.
3. Work with real-world datasets and implement advanced deep learning architectures for vision, text, and sequential data.
4. Develop, deploy, and demonstrate end-to-end machine learning applications using TensorFlow tools and libraries.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

CO1. Describe the fundamentals of machine learning and TensorFlow's role in model development.
CO2. Construct, train, and evaluate machine learning models using TensorFlow and Keras.
CO3. Design deep learning solutions (CNN, RNN, LSTM, NLP models) using TensorFlow for real-world applications.
CO4. Build and deploy complete machine learning pipelines, including model optimization and deployment using TensorFlow Lite or web apps.

Module No.	Unit No.	Contents	No of Hrs.	CO
1	**Foundations of Machine Learning and Introduction to tensorflow**		9	CO1
	1.1	Machine Learning Fundamentals: Introduction to core concepts like supervised and unsupervised learning, regression, classification, clustering, evaluation metrics.		
	1.2	Introduction to TensorFlow: Architecture and Key Concepts, Tensors, Variables, and Operations, Computational Graphs.		
	1.3	Introduction to Neural Networks: Basics of Artificial Neural Networks (ANNs), perceptrons, activation functions.		
2	Model building and training with TensorFlow		09	CO2
	2.1	Data Handling: NumPy, Pandas, CSVs, Introduction to tf. data pipelines, Feature Engineering		
	2.2	Model Architectures: Building various machine learning models (linear regression, logistic regression, etc.) with TensorFlow and Keras.		
	2.3	Training and Optimization: Gradient Descent, backpropagation, different optimizers (SGD, Adam), and techniques like regularization and mini-batching.		
	2.4	TensorBoard: Visualizing the training process and model performance.		
3	Deep learning with TensorFlow		09	CO3
	3.1	Convolutional Neural Networks (CNNs): Understanding the fundamentals, applying them to image recognition tasks (like classifying images or detecting objects), and working with real-world image datasets.		
	3.2	Recurrent Neural Networks (RNNs) and LSTMs: Exploring their usage in sequence modeling and problems like sentiment analysis, text generation, and time-series forecasting.		
	3.3	Transfer Learning: Leveraging pre-trained models and fine-tuning them for new tasks to improve efficiency and performance.		
	3.4	#Self-Learning: Scaling Up: Using advanced TensorFlow features to handle larger datasets and potentially distributed training with accelerators.		
4	Advanced topics and deployment		09	CO3

	4.1	Natural Language Processing (NLP): Tokenization, word embeddings, sequence models, sentiment analysis, and chatbots using TensorFlow.		
	4.2	Time Series Forecasting: Applying deep learning and TensorFlow to predict future trends in sequential data.		
	4.3	Deployment: Using tools like TensorFlow Lite for deploying models on mobile and embedded devices		
5	Practical projects and case studies		09	CO4
	5.1	Working on real-world problems and building end-to-end solutions using TensorFlow.		
	5.2	Case studies: Image classifiers, stock price predictors, sentiment analysis systems, and object detectors.		
		Total	45	--

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Aurélien Géron	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow	O'Reilly	2nd Edition, 2019
2	François Chollet	Deep Learning with Python	Manning Publications	2nd Edition, 2021.
3	Prateek Joshi	Artificial Intelligence with Python	Packt Publishing	2nd Edition, 2020
4	Ian Goodfellow, Yoshua Bengio, Aaron Courville	Deep Learning	MIT Press	2016
5	Nikhil Buduma & Nicholas Locascio	Fundamentals of Deep Learning	O'Reilly Media	2017
6	https://www.tensorflow.org/guide/keras			
7	NPTEL Course: Deep Learning https://onlinecourses.nptel.ac.in/noc21_cs05/preview			

Course Code	Course Title
216U01P801	**PROJECT – III**

	TH	P	TUT	Total
Teaching Scheme(Hrs.)	--	14	--	14
Credits Assigned	--	07	--	07

Examination Scheme	Marks							
	CA		ESE	TW	O	P	LabCA	Total
	ISE	IA						
	--	--	--	--	--	--	200	200

For Students opted for one semester long internship for semester VII, Project I of Semester VI will continue to Project II in Semester VIII.

For Students opting for one semester long internship for semester VIII, Project II will not continue to Project III.

Course Outcomes:

After successful completion of the course the student will be able to :

CO1. Research Paper writing and publication / participation in project competition on the Project Topic.
CO2. Suggest implementation for the identified future scope development.
CO3. Write detailed Project report
CO4. Presentation and Demonstration of the work done.
CO5. Demonstrate individual, group and leadership skills during project execution

Term work:

This is the extension of Project II under taken in semester VII. The students are required to report to the faculty advisor about the progress of the work on regular basis. There will be an intermediate evaluation at the mid of the semester by a panel of examiners. A continuous assessment record of the progress of the project will be maintained by concerned faculty advisor.

The students will be required to publish and/or present a research paper based on the work they have undertaken. They will be encouraged to participate in project competitions.

The final evaluation will be done by a pair of external examiner and internal examiner. The evaluation will include; a presentation by the group members on the project along with demonstration of the work done and oral examination. In the examination each individual student will be assessed for his/her contribution, understanding and knowledge gained.